

E524.08 / E524.09

RoHS 
compliant

General Description

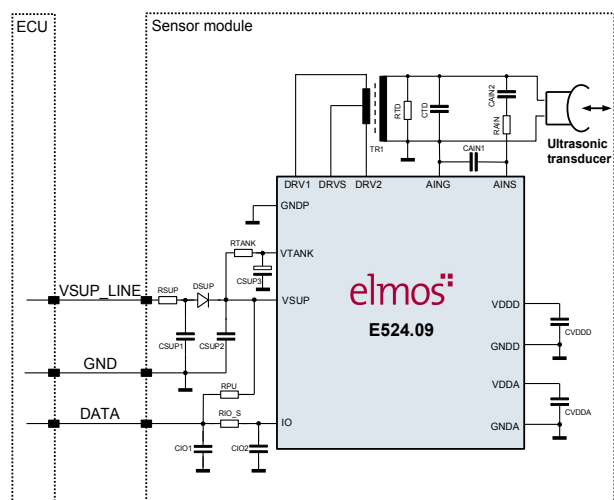
- The device builds the core for a robust and easy-to-handle distance measurement system, while offering flexibility for customer applications.

The received echo signal is amplified, converted and digitally processed. Customized obstacle interpretation is feasible by a variable detection threshold. A flexible IO protocol combined with STC (Sensitivity time control), ATG (Automatic threshold generation), NFTG (Near-field threshold generation), FTC algorithm and EPD (Echo peak detection) optimize short and long range performance.

Applications

- ## Ordering Information

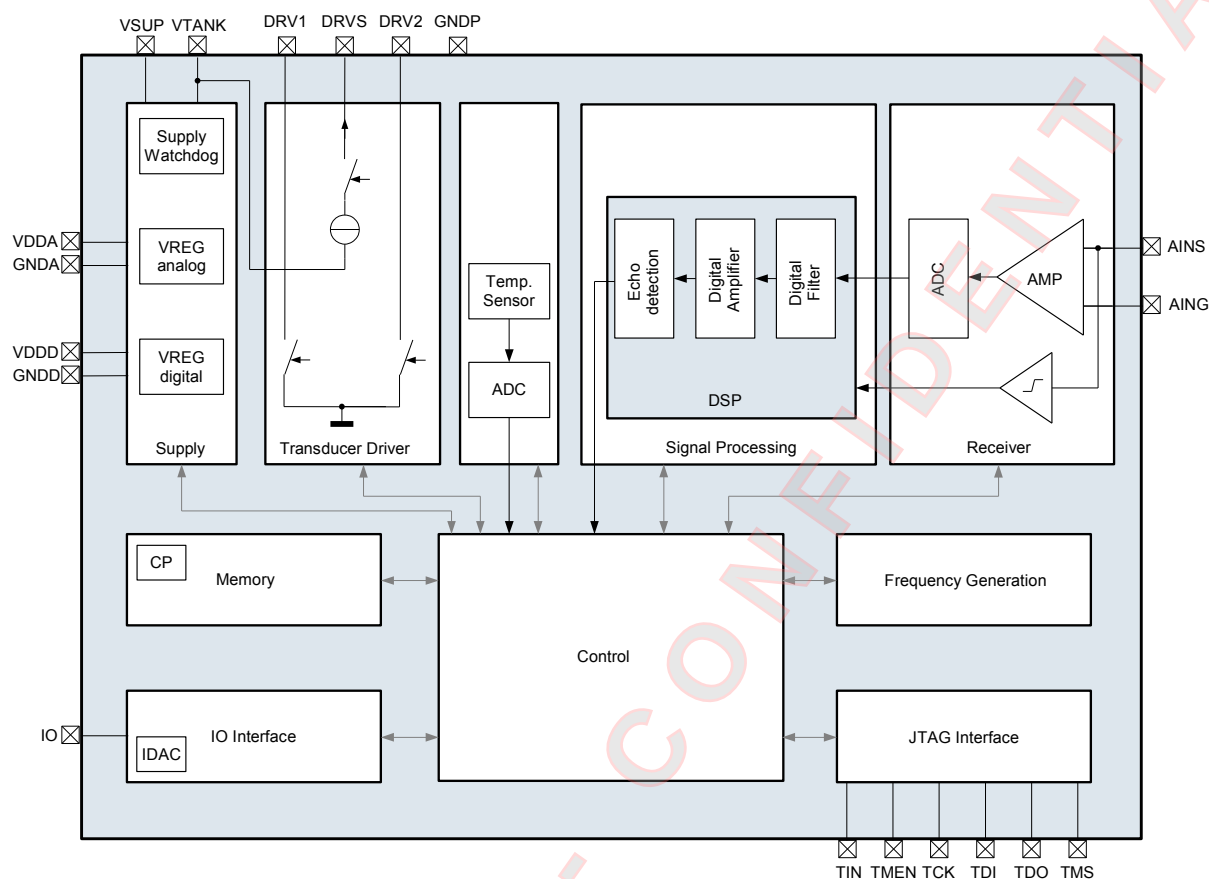
Product ID	Temp. Range	Interface	Package
E524.08	-40°C to +105°C	2-wire	QFN20L4
E524.09	-40°C to +105°C	3-wire	QFN20L4



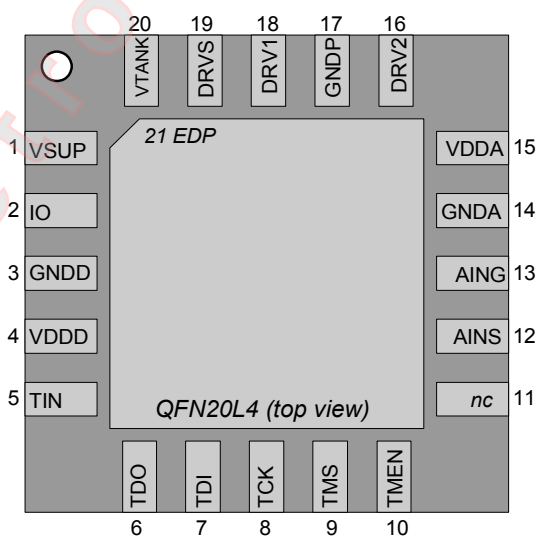
Elmos Semiconductor AG

QM-No.: 25DS0138E.00

Functional Diagram



Pin Configuration



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Pin Description

No	Name	Type	Description
1	VSUP	S	Supply voltage
2	IO	HV_A_IO	Bidirectional interface
3	GNDD	S	Digital ground
4	VDDD	A_O	Internal digital supply voltage
5	TIN	A_I	JTAG test input, should be connected to PCB-GND in application
6	TDO	D_O	JTAG data output, should be connected to PCB-GND in application
7	TDI	D_I	JTAG data input, should be connected to PCB-GND in application
8	TCK	D_I	JTAG clock, should be connected to PCB-GND in application
9	TMS	D_I	JTAG test mode select, should be connected to PCB-GND in application
10	TMEN	D_I	Test mode enable, should be connected to PCB-GND in application
11	nc		not connected to device, should be connected to PCB-GND
12	AINS	A_I	Positive signal input
13	AING	A_I	Negative signal input
14	GNDA	S	Analog ground
15	VDDA	A_O	Internal analog supply voltage
16	DRV2	HV_A_O	Driver output 2
17	GNDP	S	Power ground
18	DRV1	HV_A_O	Driver output 1
19	DRVS	HV_A_O	Transformer center tap
20	VTANK	S	Supply for transformer
21	EDP	S	Exposed die pad, has to be connected to PCB-GND

Note: A = Analog, D = Digital, S = Supply, I = Input, O = Output, B = Bidirectional, HV = High Voltage

1 Absolute Maximum Ratings

Stresses beyond these absolute maximum ratings listed below may cause permanent damage to the device. **These are stress ratings only; operation of the device at these or any other conditions beyond those listed in the operational sections of this document is not implied.** Exposure to absolute maximum rated conditions for extended periods may affect device reliability. All voltages referred to VGND. Currents flowing into terminals are positive, those drawn out of a terminal are negative.

Table 1-1: Absolute Maximum Ratings Table

No.	Description	Condition	Symbol	Min	Max	Unit
1	Supply voltage		VSUP	-0.3	36	V
2	Supply voltage	t < 500 ms	VSUP	-0.3	40	V
3	Voltage at pin IO		V _{IO}	-0.3	36	V
4	Voltage at pin IO	t < 500 ms	V _{IO}	-0.3	40	V
5	Voltage at digital pins (TCK, TDO, TDI, TMS, TMEN) and TIN		V _D	-0.3	3.6	V
6	Voltage at analog pins (AINS, AING)		V _A	-1.5	1.5	V
7	Internal digital supply voltage		VDDD	-0.3	3.6	V
8	Internal analog supply voltage		VDDA	-0.3	3.6	V
9	Voltage at pins DRV1, DRV2, DRVS and VTANK		V _{DRV1} , V _{DRV2} , V _{DRVS} , V _{TANK}	-0.3	40	V
10	Ambient temperature		T _{AMB}	-40	105	°C
11	Junction temperature		T _J	-40	125	°C
12	Storage temperature soldered	without tape, soldered to PCB	T _{STG,soldered}	-40	125	°C
13	Storage temperature unsoldered in tape	unsoldered	T _{STG,unsoldered}	-40	35	°C

2 ESD and Latch-up

2.1 ESD Protection

Table 2.1-1: ESD Conditions

Description	Condition	Symbol	Min	Max	Unit
ESD HBM protection at pins VSUP and IO	1)	$V_{\text{ESD(HBM)}}$	-4	4	kV
ESD HBM protection at all other pins	1)	$V_{\text{ESD(HBM)}}$	-2	2	kV
ESD CDM protection at pins near corners (Pins 1,5,6,10,11,15,16,20)	2)	$V_{\text{ESD(CDM)C}}$	-750	750	V
ESD CDM protection at all other pins	2)	$V_{\text{ESD(CDM)}}$	-500	500	V

1) According to AEC-Q100-002 (HBM) chip level test

2) According to AEC-Q100-011 (CDM) chip level test

2.2 Latch-up

Latch-up performance is validated according JEDEC standard JESD 78 in its valid revision.

3 Recommended Operating Conditions

Table 3-1: Recommended Operating Conditions

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Supply voltage		VSUP	6		18	V
2	EEPROM programming voltage at pin IO		V _{PROG}	23	25	27	V
3	DC input voltage at pins AINS and AING		V _{AINS} /V _{AING}	-1.0		1.0	V
4	Current into pins AINS and AING		I _{AINS} /I _{AING}	-20		20	mA
5	Transducer driver frequency		f _{DRV}	38		72	kHz

4 Electrical Characteristics

- $V_{SUP} = +6V$ to $+18V$,
- $T_{AMB} = -40^{\circ}C$ to $+105^{\circ}C$, unless otherwise noted.
- Typical values are at $V_{SUP} = 12V$ and $T_{AMB} = +25^{\circ}C$.
- Positive currents flow into the device pins.
- All time values are based on the typical oscillator frequency $f_{OSC,typ}$.

4.1 Power Supply

Table 4.1-1: Supply and POR - Electrical Parameters

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Current consumption at VSUP	$t > t_{D_nom}$	I_{VSUP}	-	4.5		mA
2	Current consumption at VSUP in standby mode		I_{VSUP_STBY}	-	-	1	mA
3	Internal generated analog supply	external C_{VDDA} connected	VDDA	3.1	3.3	3.5	V
4	Current out of VDDA in case of short to GND		I_{VDDA_SHORT}	-55	-	-	mA
5	Internal generated digital supply	external C_{VDDD} connected	VDDD	3.1	3.3	3.5	V
6	Current out of VDDD in case of short to GND		I_{VDDD_SHORT}	-55	-	-	mA
7	Level at VSUP: threshold from operation mode to reset state		$VSUP_{POR}$	-	-	4.5	V
8	Start-up time until communication via IO is possible ^{*)}		t_{D_IO}	-	260	800	μs
9	Delay time until full functionality after wake up ^{*)}	$VSUP \geq 10V$	t_{D_WAKEUP}	-	-	2	ms
10	VSUP undervoltage detection low level		$VSUP_{UV_L}$	4.7	5.0	5.3	V
11	VSUP overvoltage detection high level		$VSUP_{OV_H}$	19.5	21	22.5	V

^{*)} Not tested in production

4.2 Frequency generation

Table 4.2-1: Oscillator - Electrical Parameters

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Oscillator frequency	@ $T_{AMB} = 25^{\circ}C$	f_{OSC}	11.94	12.0	12.06	MHz
2	Temperature coefficient of oscillator ^{*)}		TC_{f_OSC}	-450	-300	-150	ppm/K

^{*)} Not tested in production

4.3 Transducer driver

Table 4.3-1: Transducer driver - Electrical Parameters

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Lowest selectable driver current	I_DRV=0 V_DRV=2.5V	I_DRV,0	-	168	-	mA
2	Highest selectable driver current	I_DRV=31 V_DRV=2.5V	I_DRV,31	-	354	-	mA
3	Number of steps for driver current adjustment		N_IDRV	-	31	-	-
4	Increment of driver current		I_DRV,INC	-	6	-	mA
5	Pull down resistor at pin DRVS	V_DRV=0.2V Receive Mode	R_PD,DRVS	2	5	15	kΩ
6	On resistance of driver switches SW_DRVX	Burst Mode	R_SW,DRV1,2	-	1.2	2	Ω
7	Transducer driver frequency	to be adjusted by F_DRV	f_DRV	1/144	-	1/399	f _{osc}

4.4 Receiver

Table 4.4-1: Analog Amplifier - Electrical Parameters

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Minimum gain of analog input amplifier	@ f_DRV=58 kHz	G_ANA_MIN	33	36	39	dB
2	Maximum gain of analog input amplifier	@ f_DRV=58 kHz	G_ANA_MAX	47	50	53	dB
3	Step size of analog gain ¹⁾	@ f_DRV=58 kHz	ΔG_ANA	-	2	-	dB
4	Number of steps		N_G_ANA	-	7	-	
5	Input Impedance AINS to GNDA AING to GNDA	V_AINS≤0.1 V, V_AING≤0.1 V	R_IN	75	100	125	kΩ
6	Noise Level ¹⁾	@ f_DRV=58 kHz / max gain	e _N	-	3.8	-	$\frac{nV}{\sqrt{Hz}}$

¹⁾ Not tested in production

¹⁾ monotony guaranteed by design

4.5 Signal Processing

Table 4.5-1: Digital Filter - Electrical Parameters

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Center frequency		$f_{c, \text{filt}}$	-	1	-	f_{DRV}
2	-3dB-bandwidth	automatically selected with respect to NPULSES and FILTER_CFG	$f_{-3\text{dB}}$	0.0258	-	0.219	f_{DRV}

Table 4.5-2: Digital Amplifier - Electrical Parameters

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Digital gain	to be adjusted by G_DIG	G_{DIG}	0	-	47.89	dB
2	Step size of digital gain		ΔG_{DIG}	0.28	0.38	0.53	dB
3	Number of digital gain steps		N_{G_DIG}	-	127	-	

Table 4.5-3: STC - Electrical Parameters

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Start time of digital gain increment after any measurement command	to be adjusted by STC_START	$T_{\text{STC_START}}$	0.874	-	1.574	ms
2	Ramp up gain of STC	to be adjusted by STC_CFG	G_{STC}	0	-	44.91	dB

4.6 Temperature sensor

Table 4.6-1: Temperature sensor - Electrical Parameters

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Temperature sensor code at 25°C		TSENS_{C_RT}	413	433	453	LSB
2	Accuracy of temperature sensor output ^{*)}		$\text{TSENS}_{\text{ACC}}$	-10	-	10	°C
3	Slope of temperature sensor ^{*)}		$\text{TSENS}_{\text{SLOPE}}$	-	0.491	-	°C/LSB
4	Integration time of ADC ^{*)}		TSENS_{T_DEL}	-	5.5	-	ms

^{*)} Not tested in production

4.7 IO Interface

Table 4.7-1: IO Interface (DC) - Electrical Parameters

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Voltage threshold for pin IO to detect low input signal		V_{IO_IL}	0.29	0.33	0.36	VSUP
2	Voltage threshold for pin IO to detect high input signal		V_{IO_IH}	0.62	0.67	0.70	VSUP
3	Voltage at pin IO to output low level at high current load	$I_{OUT} = 16\text{mA}$ [only 524.08]	$V_{IO_OL_16}$	-	-	0.8	V
4	Voltage at pin IO to output low level	$I_{OUT} = 4\text{mA}$	$V_{IO_OL_4}$	-	-	0.5	V
5	Current limitation for pin IO	$V_{OUT} < 18\text{V}$	I_{IO_LIMIT}	-	-	60	mA
6	Resolution of current DAC for envelope via IO ^{*)}		N_{IDAC}	-	6	-	Bit
7	Current per LSB for envelope via IO		I_{IDAC_STEP}	-	16.6	-	$\mu\text{A}/\text{LSB}$
8	Maximum IDAC output current for envelope via IO		I_{IDAC_max}	900	1060	1250	μA

^{*)} Not tested in production

Table 4.7-2: IO Interface (AC) - Electrical Parameters

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Slew Rate		SR_{VIO}	-	1.7	-	$\text{V}/\mu\text{s}$
2	Input debouncer		T_{DEB}	0	-	20	μs
3	IO high phase after T_{CMD}		T_D	21	50	71.2	μs
4	IO low phase for send request		T_{SND}	78.8	100	118.7	μs
5	IO low phase for receive request		T_{REC}	131.3	150	166.2	μs
6	IO low phase beginning of measurement command		T_{MEAS}	183.8	200	213.7	μs
7	IO low phase to enter command mode		T_{CMD}	236.3	250	261.3	μs
8	Setup time for V_{PROG}		T_{VPROG}	-	-	5	ms
9	Programming time		T_{PROG}	22	25	-	ms
10	Bit length, ECU to sensor module		T_{BIT_WR}	131.3	150	166.2	μs
11	IO low phase for a logical '0', ECU to sensor module		T_{BIT0_WR}	78.8	100	118.7	μs
12	IO low phase for a logical '1', ECU to sensor module		T_{BIT1_WR}	21	50	71.2	μs
13	Bit length, sensor module to ECU		T_{BIT}	142.5	150	157.5	μs
14	IO low phase for a logical '0', sensor module to ECU		T_{BIT0}	95	100	105	μs
15	IO low phase for a logical '1', sensor module to ECU		T_{BIT1}	47.5	50	52.5	μs
16	Debouncing of echo signal		T_{DEB_ECHO}	-	3	-	$1/f_{DRV}$
17	Pulse width echo event in 2 wire mode	[only 524.08]	$T_{ECHO2WIRE}$	47.5	50	52.5	μs
18	Pulse width low frequency calibration pulse		T_{CAL}	-	496	-	$1/f_{DRV}$
19	Pulse width of calibration pulses		T_{CALS}	-	8	-	$1/f_{DRV}$

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Table 4.7-3: Eeprom Bits - Delivery State

No.	Description	Condition	Symbol	Min	Typ	Max	Unit
1	Customer bits for free use	free configurable by customer	CUSTOMER BITS	0	-	0	-
2	Digital amplifier gain configuration		G_DIG	8	-	8	-
3	Analog amplifier gain configuration		G_ANA	4	-	4	-
4	Transducer output current configuration		I_DRV	16	-	16	-
5	Driver frequency configuration		F_DRV	63	-	63	-

5 Functional Description

5.1 Overview

5.1.1 Ultrasonic distance measurement

The principle of ultrasonic distance measuring is based on transmitting a pulse and measuring the reflection time of the received pulse as shown in Figure 5.1.1-1. The relationship between the distance from the transducer to the object L and the time T it takes to receive the echo is $L = c * T/2$, where c is the velocity of sound ($c = 343 \text{ m/s}$ at 20°C). The IC requires only one transducer, which acts as a transmitter and receives the reflected pulse with a time delay. The time delay is proportional to the distance to measure.

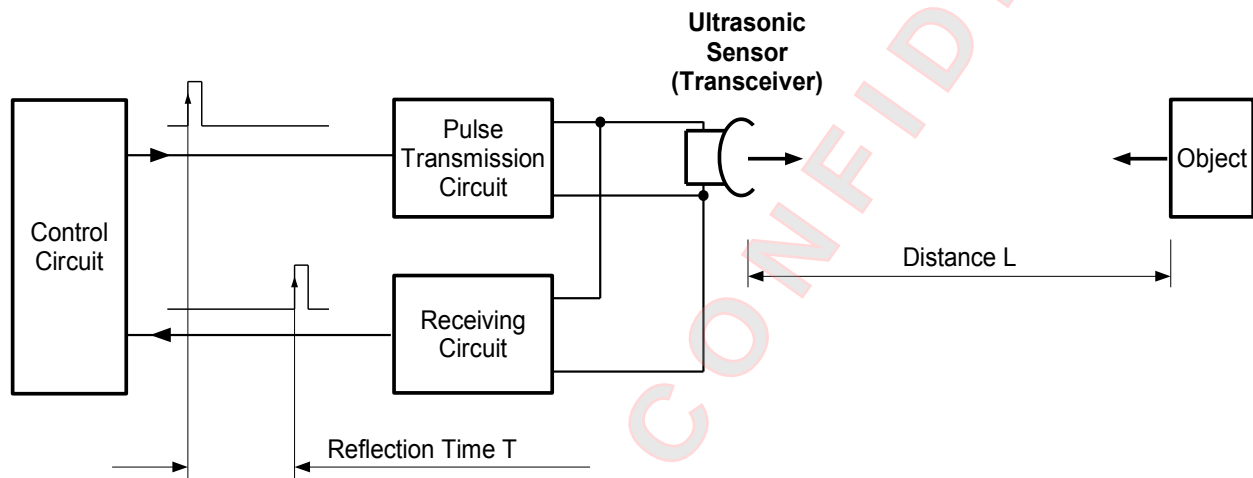


Figure 5.1.1-1: Principle: Ultrasonic distance measurement

5.1.2 Measurement cycle

An ultrasonic measurement cycle is defined as the time period, where the IC performs actions with the aim to gather an associated set of data for further processing in the control unit (ECU). A cycle may contain:

- Noise measurement
- Bursting
- Envelope data sampling
- Obstacle detection
- Collecting diagnostic information

The measurement cycle is started by a 'SEND' or 'RECEIVE' request by the control unit. Internal logic generates a burst signal for the transducer first. Incoming reflected pulses are amplified, digitally filtered and compared to customer programmable threshold levels or to the automatically generated threshold. If the echo pulse exceeds the set threshold level, the IC triggers the IO line. The time elapsed from sending out a burst signal to receiving an echo signal from an object is proportional to the distance of the object.

Once a cycle is initiated by the control unit, it cannot be interrupted during the programmable measurement time. If no measurement is performed the device is in Idle state.

5.1.3 Measurement modes

The device supports the following modes of operation:

- **SEND mode (Direct mode):** Bursting and measuring
- **RECEIVE mode (Indirect mode):** No bursting, only measuring

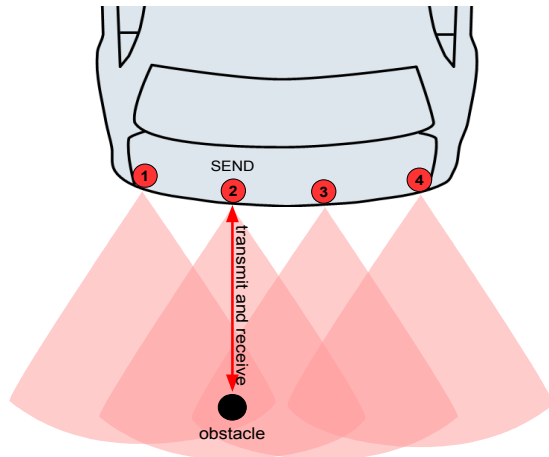


Figure 5.1.3-1: Direct Measurement

In **SEND mode** (direct measurement) the sensor transmits and receives its own ultrasonic signal.

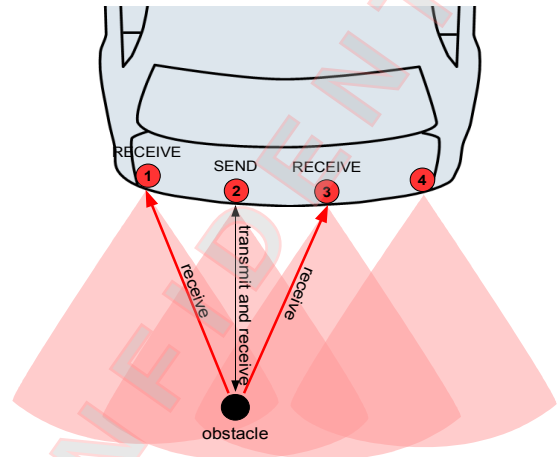


Figure 5.1.3-2: Indirect Measurement

In **RECEIVE mode** (indirect measurement) the sensor does not send any ultrasonic signal. It only receives a signal send out by another sensor.

5.1.4 Interface

The communication between the control unit and the sensor IC is possible via 2-wire or 3-wire interface. The E524.08 supports a bi-directional communication via data modulation and the E524.09 supports a dedicated IO-line for data transfer.

E524.08

The E524.08 requires only two wires for the communication of ultrasonic module and ECU. Ground (GND) and the supply wire (VSUP_DATA). As illustrated in Figure 7.1-1, bidirectional data travels between ultrasonic module and ECU by modulating voltage on the supply wire.

E524.09

The E524.09 requires three wires to communicate from the ultrasonic module with the ECU. As shown in Figure 7.3-1, the three wires are ground (GND), supply (VSUP) and one signal line (DATA).

5.2 Power Supply

The IC uses two internal regulators (VDDA and VDDD) as supplies for the internal circuits.

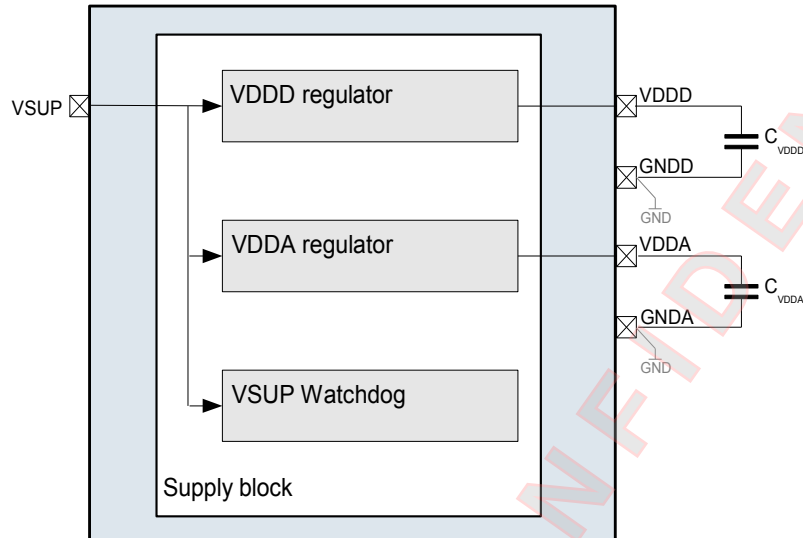


Figure 5.2-1: Supply block

5.2.1 Analog supply

The analog supply VDDA requires an external blocking capacitor C_{VDDA} . It is recommended to place the external capacitor as close as possible to the related pins VDDA and GNDA. VDDA is not intended to supply any external components. In case VDDA is shorted to GND, the output current will be limited to $I_{VDDA,SHORT}$.

5.2.2 Digital supply

The digital supply VDDD requires an external blocking capacitor C_{VDDD} . It is recommended to place the external capacitor as close as possible to the related pins VDDD and GNDD. VDDD is not intended to supply any external components. In case VDDD is shorted to GND, the output current will be limited to $I_{VDDD,SHORT}$.

5.2.3 VSUP Watchdog

Two comparators monitor the supply voltage to detect overvoltage and undervoltage during a measurement.

- If VSUP rises above $VSUP_{OV,H}$ no further pulses can be sent to protect the driver structure.
- If VSUP decreases below $VSUP_{UV,L}$ no further pulses can be sent to prevent data loss due to reset below $VSUP_{POR}$.
- In both cases the status flags are set and no further burst pulses will be sent until $VSUP_{UV,H} < VSUP < VSUP_{OV,L}$.

With 'READ_STATUS' the undervoltage and overvoltage information can be read out.

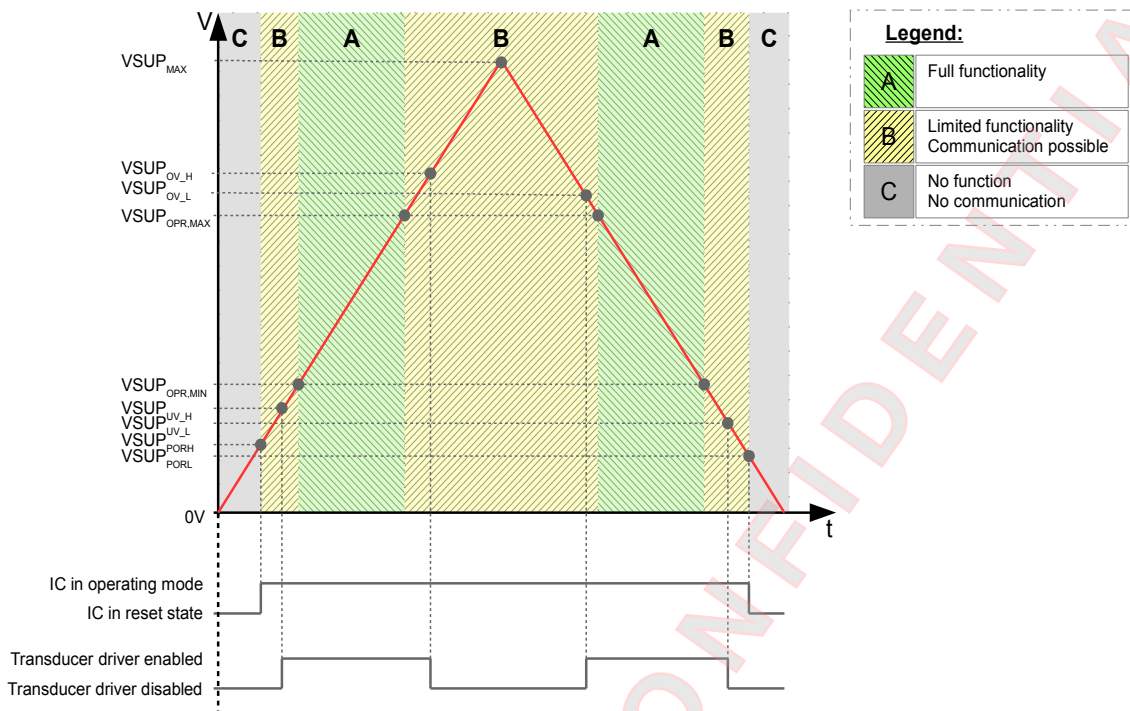


Figure 5.2.3-1: Overvoltage and undervoltage detection

The digital part is running, when the power-on-reset is high. When the power-on-reset goes from high to low state, the digital part is in reset state. This results in full functionality down to $VSUP_{min}$ and a still working digital part down to $VSUP_{POR}$. Between $VSUP_{min}$ and $VSUP_{POR}$ the performance is limited, but communication with the control unit is unrestricted possible and data retention is guaranteed.

5.2.4 Power up sequence

The IC begins to power up once an external voltage is supplied to the pin VSUP. A power-up diagram is shown in Figure 5.2.4-1. The sequence can be described as follows:

1. An external supply voltage is supplied to pin VSUP.
2. After the Power-on-reset (POR) signal to the digital part is released, the digital part is running and communication via the interface is possible. The receiver path is enabled to receive signals.

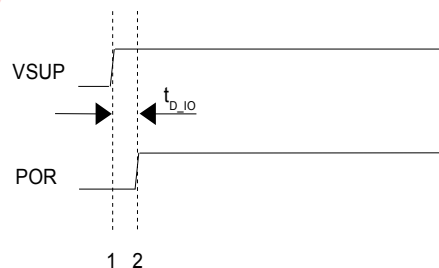


Figure 5.2.4-1: Power up sequence

5.2.5 Standby mode

This IC supports a standby mode to reduce current consumption.

- The standby mode is entered with command 'STANDBY'.
- The command 'WAKE_UP' initiates a wake up from standby mode. Full performance is available after the wait time t_{D_WAKEUP} . Any other valid command also leaves the standby mode, but full functionality is only available after 'WAKE_UP' command and wait time t_{D_WAKEUP} .

For minimum current consumption in standby mode the IO has to be at VSUP level.

5.3 Frequency generation

The frequency generation concept consists of an oscillator with a fixed frequency f_{OSC} for IO communication, threshold generation and measurement time. It has a specified temperature coefficient of $\text{TC}_{f_{\text{OSC}}}$. The oscillator frequency can be configured by OSC_TRIM, which is stored in the EEPROM memory.

The transducer driver frequency is derived from f_{OSC} with a digital divider, which can be set by F_DRV. The burst frequency configuration is explained in 5.4.1.1. Following figure shows the clocking scheme:

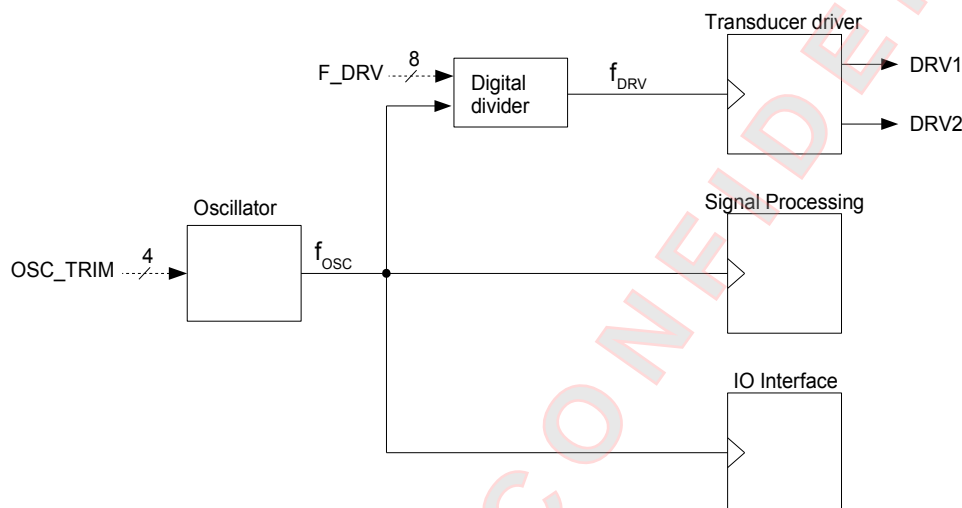


Figure 5.3-1: Block diagram of clock distribution

5.3.1 Oscillator frequency configuration

The frequency f_{OSC} can be fine adjusted by the EEPROM bits OSC_TRIM. The frequency is strictly monotonic.

With IC delivery the frequency f_{OSC} is trimmed. Due to stress during soldering this frequency may change. Therefore it is possible to trim f_{OSC} , which also affects the frequencies used within the transducer driver, signal processing and IO interface. Using OSC_TRIM the current frequency can be increased or decreased according to following table:

Table 5.3.1-1: Oscillator trimming

OSC_TRIM[3:0]	Meaning
1000	-4.0%
1001	-3.5%
...	
1111	-0.5%
0000	default - no trimming
0001	+0.5%
...	
0110	+3.0%
0111	+3.5%

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The procedure for adjusting the frequency should be as following:

1. Set F_DRV according to the desired frequency with 'CALIB_WRITE'
2. Measure the current frequency with 'CAL_PULSES'
3. Calculate the percental difference between target frequency and current frequency
4. Choose value for OSC_TRIM according to Table 5.3.1-1
5. Repeat until the target frequency is reached
6. Store the setting for OSC_TRIM by 'EE_COPY' command

5.4 Transducer driver

The transducer is powered by a push-pull center-tapped transformer for maximum efficiency. The Transducer driver builds the interface between the supply voltage VTANK and the transformer-transducer conjunction. There are two modes of operation:

- receive mode
- burst mode

The circuit below introduces the notation of the following chapters.

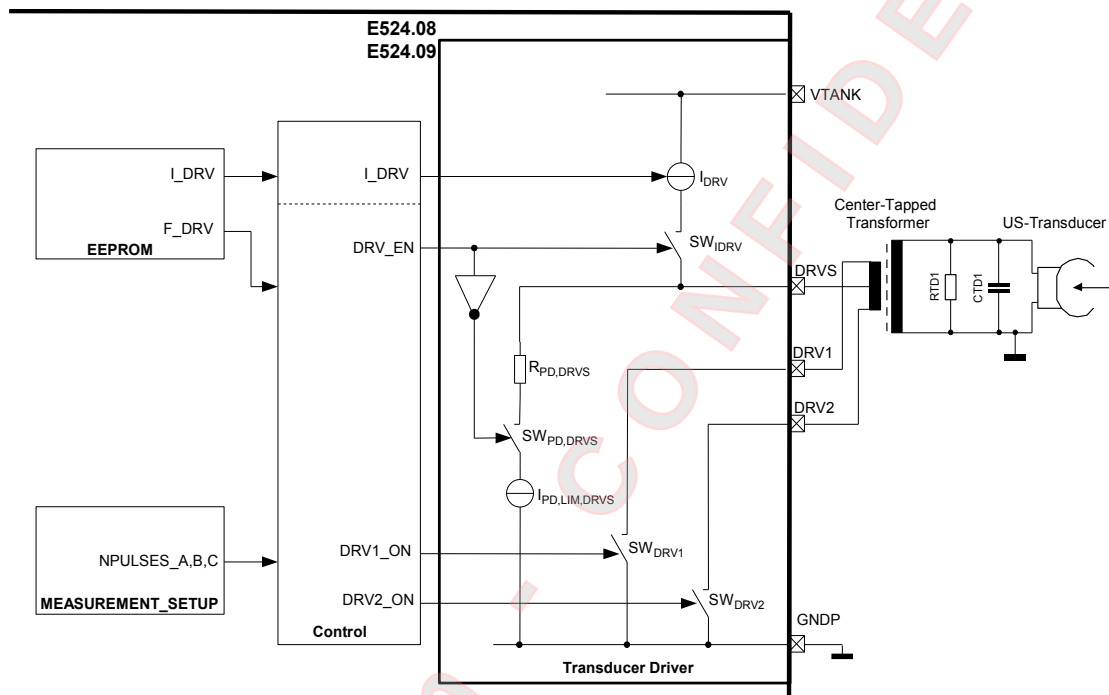


Figure 5.4-1: Transducer driver block diagram

Receive Mode

In receive mode the Transducer driver isolates the transformer-transducer conjunction from VTANK in order to prevent noise coupling from the supply to the transducer. Therefore the current source I_{DRV} is disconnected from pin DRVS by switching off SW_{IDRV} . Pin DRVS is pulled to GNDP via $R_{PD,DRVS}$ by $SW_{PD,DRVS}$. The current into pin DRVS is limited by $I_{PD,LIM,DRVS}$. Switches SW_{DRV1} and SW_{DRV2} are off so the primary winding of the transformer is only connected with the center tap.

Burst Mode

The burst mode generates a power controlled ultra sound burst signal via the external transformer and transducer. In burst mode $SW_{PD,DRVS}$ is switched off and the adjustable current source I_{DRV} is connected to pin DRVS via SW_{IDRV} , supplying the center tap of the transformer. Higher currents I_{DRV} lead to higher transducer power. The burst signal is generated by forcing the current I_{DRV} alternately into one of the transformers primary windings, which are connected to pins DRV1 and DRV2. This is known as 'push-pull principle'. A current is forced into a winding, by setting the corresponding driver switch SW_{DRV1} or SW_{DRV2} . Thus, these switches are set alternating according to the selected burst signal configuration. The switching frequency of SW_{DRV1} and SW_{DRV2} is the burst frequency f_{DRV} . In order to keep the transducer power constant even for low supply voltages VTANK, the on-resistance $R_{SW,DRVX}$ of the switches SW_{DRVX} is low. The burst signal consists of burst pulses, the length of the burst signal (burst length) is defined by the number of burst pulses. A burst pulse has the period $1/f_{DRV}$.

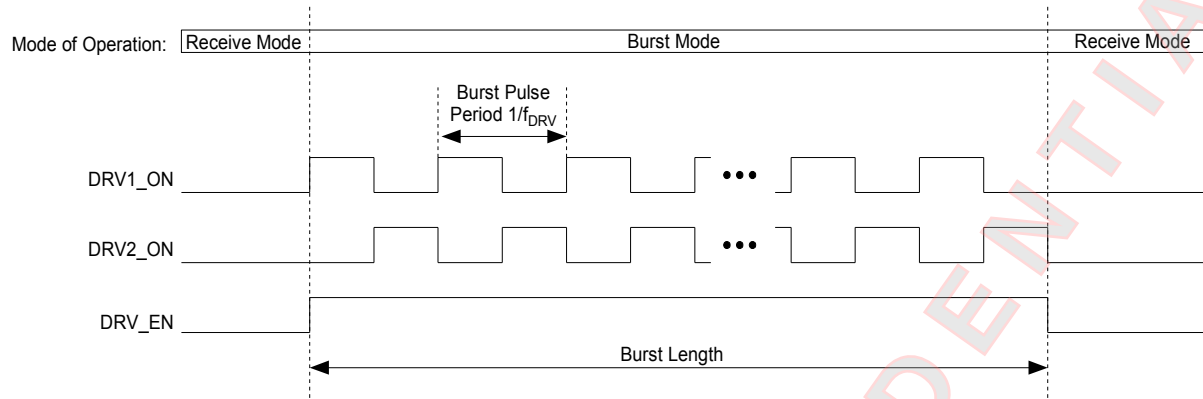


Figure 5.4-2: Timing Diagram for Transducer Driver

5.4.1 Configuration

The ultrasonic burst signal can be configured by:

- burst frequency F_{DRV}
- burst power by definition of the driver current I_{DRV}
- burst length by definition of NPULSES_X (X depends on the selected measurement profile)

Parameters F_{DRV} and I_{DRV} are stored in the EEPROM memory, while the configuration of NPULSES_X is stored in the volatile memory of MEAS_SETUP register (see Table 5.9.4-1).

5.4.1.1 Burst frequency configuration

The burst frequency f_{DRV} can be adjusted by the EEPROM bits F_{DRV} and is calculated by:

$$F_{DRV} = \frac{f_{OSC}}{f_{DRV}} - 144$$

The burst frequency is illustrated in Figure 5.4.1.1-1

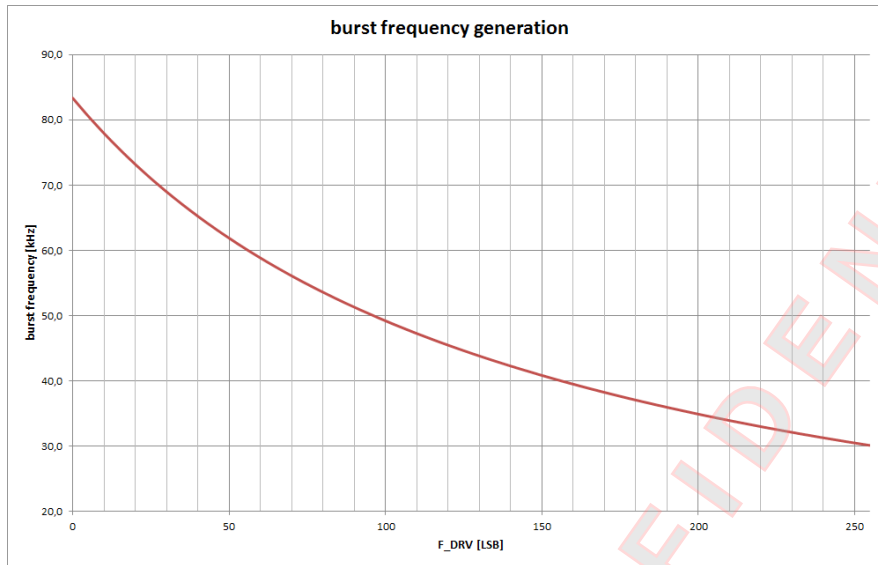


Figure 5.4.1.1-1: Burst frequency generation

The procedure for adjusting the frequency should be as following:

1. Select the frequency divider
2. Write F_DRV to the EEPROM

5.4.1.2 Driver current configuration

The driver current I_{DRV} can be adjusted according to the system and environmental conditions by EEPROM register bits I_DRV.

- The minimum selectable I_{DRV} current is specified by parameter $I_{DRV,0}$ for I_DRV=0
- I_{DRV} can be increased from $I_{DRV,0}$ with N_{DRV} steps
- The increment for a single step is given by parameter $I_{DRV,INC}$
- The maximum selectable I_{DRV} current is specified by parameter $I_{DRV,31}$ for I_DRV=31

The adjusted driver current I_{DRV} according to the selected step I_DRV can be calculated by following formula:

$$I_{DRV} = I_{DRV,0} + I_{DRV,INC} \cdot I_DRV$$

5.4.1.3 Burst length configuration

The number of burst pulses is defined by the parameters NPULSES_A, NPULSES_B and NPULSES_C. In general an ultrasonic burst with less pulses shows a better performance in the short range while a burst with more pulses has a better performance in the long range. The burst length can be configured between 4 and 32 pulses in steps of 4 pulses.

The following table shows the burst length configuration:

Table 5.4.1.3-1: NPULSES

NPULSES[2]	NPULSES[1]	NPULSES[0]	Number of pulses
0	0	0	4
0	0	1	8
0	1	0	12
0	1	1	16
1	0	0	20
1	0	1	24
1	1	0	28
1	1	1	32

5.5 Receiver

The receiving path including receiver and signal processing block is shown in Figure 5.5-1.

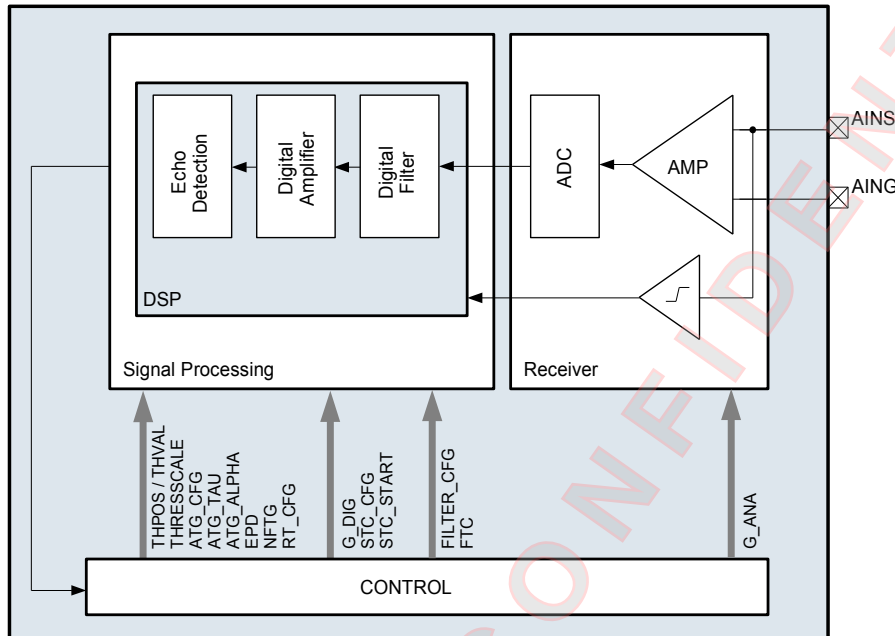


Figure 5.5-1: Block diagram of Receiver and Signal Processing

The echo signal is picked up from the transducer at pins AINS and AING. The signal goes through a programmable analog gain stage first. After digitization, the signal is further processed by a digital filter. This reduces the sensitivity to unwanted signals outside the frequency band of the transducer. A digital gain stage is used to perform further amplification before the echo detection takes place.

5.5.1 Analog Amplifier

Transducer output signal levels vary by type and application. The differential input of the amplifier minimizes pickup of unwanted signals. AINS input is typically used as signal input while AING input has to be connected to transducer GND.

The low incoming analog echo signal has to be amplified before it is converted by an ADC. The analog gain can be adjusted by G_{ANA} as follows:

$$G_{ANA} = G_{ANA_MIN} + G_{ANA} \cdot \Delta G_{ANA}$$

The analog amplification G_{ANA} can be used as coarse pre-calibration while the wide range digital amplification G_{DIG} is intended to do a fine calibration.

G_{ANA} is stored in the EEPROM memory.

5.5.2 Analog to Digital Conversion (ADC)

The analog to digital converter converts the analog signal into a digital signal and all further steps are processed digitally.

5.6 Signal Processing

5.6.1 Digital Filter

Since the clock of the transducer driver and the ADC / filter clock are synchronized, the digital filter follows the sending frequency. The output of the digital matched filter (DMF) represents the envelope of the echo signal.

The driver frequency and the frequency of the digital filter depend on the sending frequency (f_{DRV}). The center frequency of the filter is shifted with the driver frequency temperature dependency. The bandwidth is scaled with frequency. It is automatically set according to the number of pulses and can be selected as standard or wide using FILTER_CFG. FILTER_CFG is stored in the volatile memory of the MEAS_SETUP register.

The following tables and figures show the details:

Table 5.6.1-1: Standard -3dB-bandwidth of digital filter (FILTER_CFG='0')

Number of burst pulses	f_{-3dB}
4	$0.1058 * f_{\text{DRV}}$
8, 12, 16	$0.0520 * f_{\text{DRV}}$
20, 24, 28, 32	$0.0258 * f_{\text{DRV}}$

Table 5.6.1-2: Wide -3dB-bandwidth of digital filter (FILTER_CFG='1')

Number of burst pulses	f_{-3dB}
4	$0.2190 * f_{\text{DRV}}$
8, 12, 16	$0.1058 * f_{\text{DRV}}$
20, 24, 28, 32	$0.0520 * f_{\text{DRV}}$

5.6.1.1 Filter characteristics

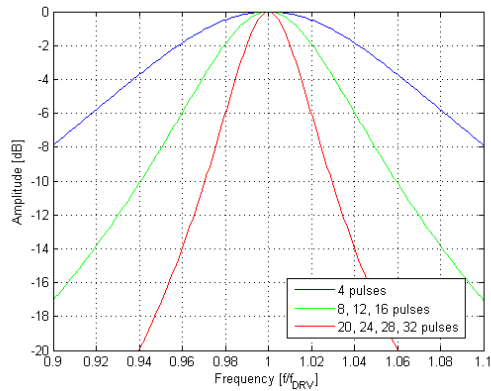


Figure 5.6.1.1-1: Normalized filter frequency response (std)

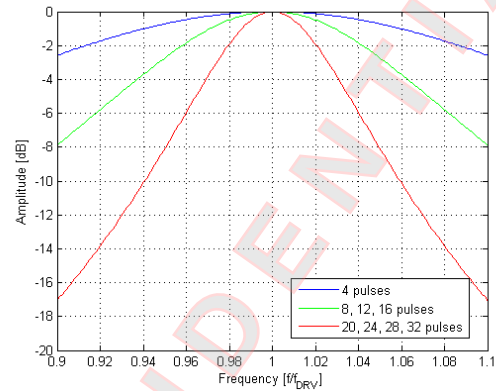


Figure 5.6.1.1-2: Normalized filter frequency response (wide)

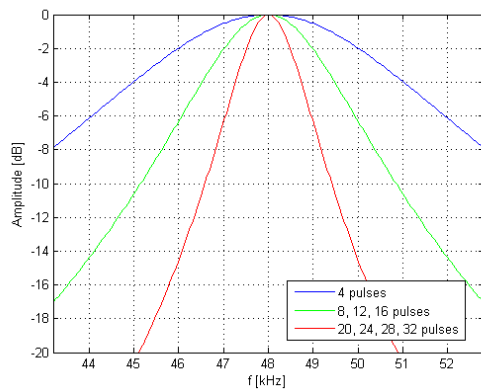


Figure 5.6.1.1-3: Filter frequency response at 48kHz (std)

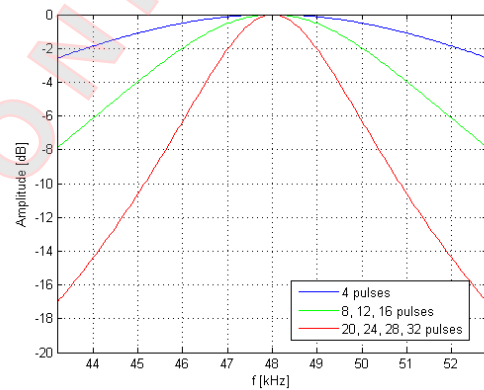


Figure 5.6.1.1-4: Filter frequency response at 48kHz (wide)

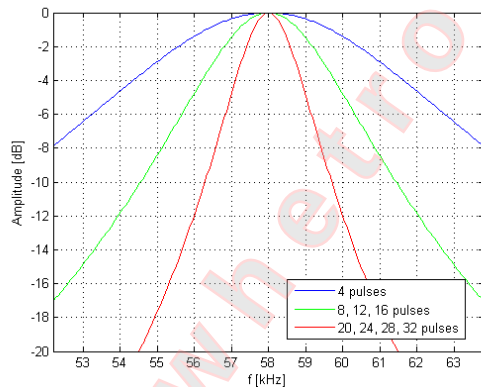


Figure 5.6.1.1-5: Filter frequency response at 58kHz (std)

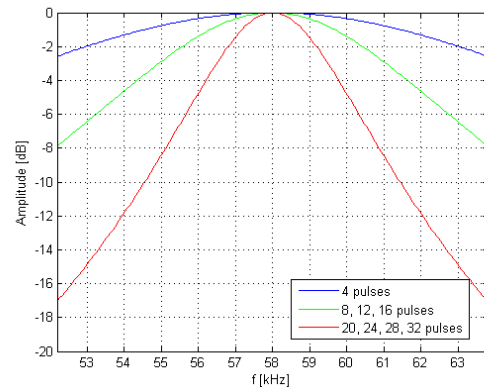


Figure 5.6.1.1-6: Filter frequency response at 58kHz (wide)

5.6.2 Digital Amplifier and Sensitivity Time Control (STC)

The digital amplifier basically performs two functions:

- a configurable digital basic amplification using G_{DIG}
- a stepwise incrementation of digital gain over time called sensitivity time control (STC)

With the following figure the notation of digital gain behavior is introduced:

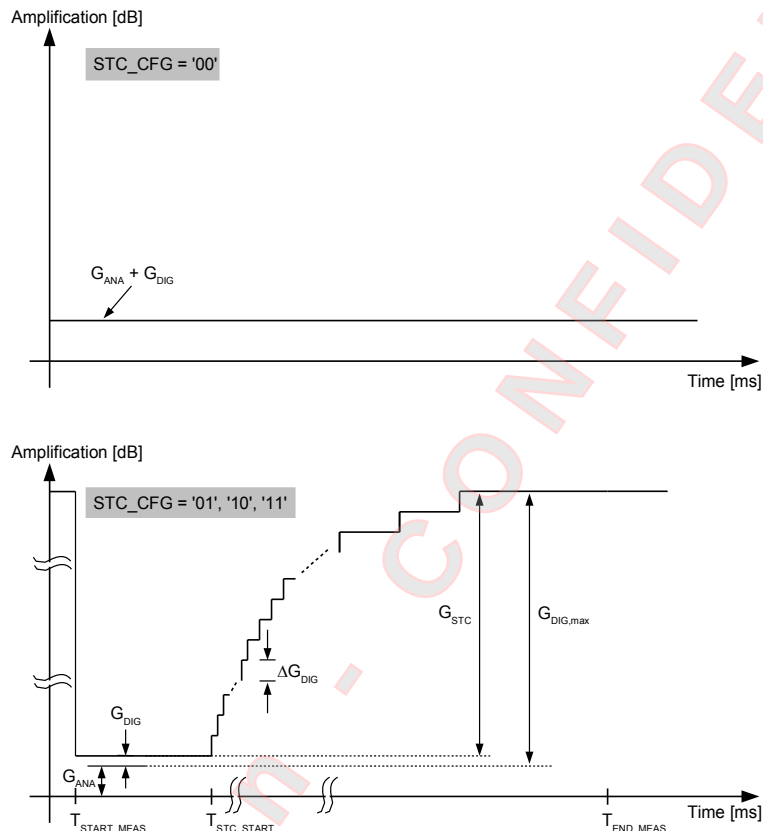


Figure 5.6.2-1: Digital gain behavior for Sensitivity Time Control

The digital gain initial value G_{DIG} can be adjusted in ΔG_{DIG} steps with G_{DIG} according to the formula:

$$G_{DIG} = G_{DIG} \cdot \Delta G_{DIG}$$

G_{DIG} is stored in the EEPROM memory.

If the STC function is not used, the amplification of the receiving path stays constant for the complete measurement cycle. The overall gain in this case is $G_{ANA} + G_{DIG}$.

Sensitivity time control (STC) increases the digital gain G_{DIG} over time until the maximum value $G_{\text{DIG,max}}$ is achieved. The incrementation begins at a start time $T_{\text{STC_START}}$ after any measurement command with an initial value of $G_{\text{ANA}} + G_{\text{DIG}}$. G_{DIG} can be configured by G_{DIG} . The ramp up start time $T_{\text{STC_START}}$ can be configured by STC_START .

The digital amplification is incremented up to a configurable maximum value with $\text{STC_CFG}='01'$, $'10'$ or $'11'$. Figure 5.6.2 shows the digital gain behavior. The maximum amplification value depends on the $\text{STC_CFG}[1:0]$ and on TMEAS . A measurement time TMEAS shorter than the time needed to ramp up the STC completely, results in a lower final gain value at the end of measurement.

The digital gain increases with an incrementation algorithm to obtain constant echo heights within a wide distance range. This algorithm increments the digital gain G_{DIG} with an increasing time step. The time step starts with T_{STEP} and doubles each 16 steps.

The STC can be activated by STC_CFG .

Table 5.6.2-1: STC_CFG

$\text{STC_CFG}[1]$	$\text{STC_CFG}[0]$	Function
0	0	STC ramps up for $G_{\text{STC}} = 0\text{dB}$ (STC off)
0	1	STC ramps up for $G_{\text{STC}} = 24\text{dB}$
1	0	STC ramps up for $G_{\text{STC}} = 30\text{dB}$
1	1	STC ramps up for $G_{\text{STC}} = \text{maximum value}$

$\text{STC_CFG}='00'$ results in a constant gain over the complete range composed of the analog gain G_{ANA} and the digital gain G_{DIG} . $\text{STC_CFG}='01'$, $'10'$ or $'11'$ result in an increasing gain over time with the possibility to adjust the initial gain value, the final gain value and the starting time of the incrementation with regards to the transducer and the overall system.

We recommend to use STC in combination with the automatic threshold generation (ATG).

The STC start time can be configured by STC_START .

Table 5.6.2-2: STC_START

$\text{STC_START}[1]$	$\text{STC_START}[0]$	$T_{\text{STC_START}} [\text{ms}]$
0	0	$1.11 + T_{\text{CORR}}$
0	1	$1.40 + T_{\text{CORR}}$
1	0	$1.69 + T_{\text{CORR}}$
1	1	$1.98 + T_{\text{CORR}}$

A principle diagram of the digital gain behavior can be found in Figure 5.6.2-1.

The configuration of STC_CFG and STC_START is written in volatile memory of MEAS_SETUP register.

5.6.3 Echo detection

The echo detection is performed by comparing the echo envelope signal of the digital filter output with a threshold value. The result appears at pin IO in SEND / RECEIVE mode. If the current envelope of the echo signal goes above the threshold for a time $t > t_{\text{DEB_ECHO}}$, pin IO is pulled to GND.

The kind of threshold generation as well as the echo detection type (standard or peak detection) has high impact on the accuracy of IO-line output.

The IC offers two different possibilities of threshold generation:

- static threshold generation
- automatic threshold generation (combination of static and dynamic threshold generation)

The threshold behavior can be configured with the static threshold values and parameters ATG_CFG, ATG_TAU and ATG_ALPHA of the automatic threshold within the volatile THRES_SETUP register.

With the example for static threshold generation in Figure 5.6.3-1 the notation of the following chapters is introduced.

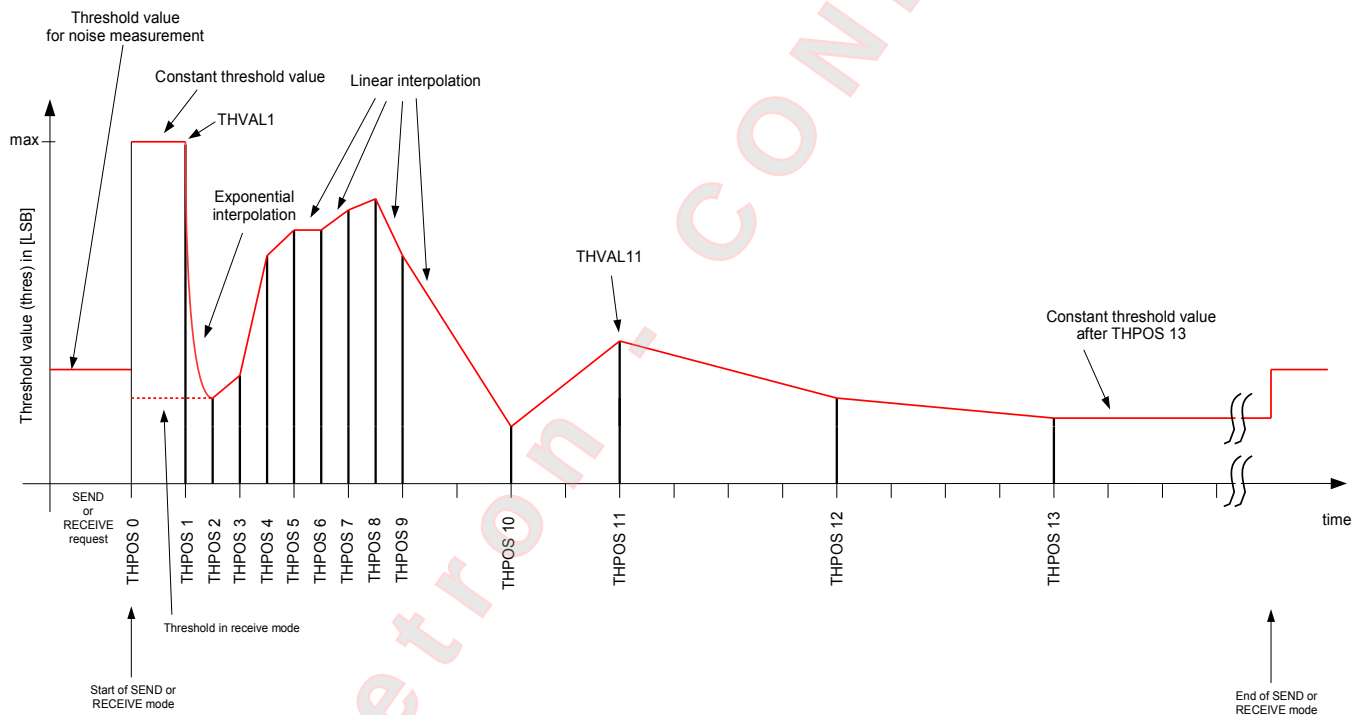


Figure 5.6.3-1: Intervals of static threshold generation in SEND or RECEIVE mode and related threshold values

5.6.3.1 Static threshold curve

Each threshold point of the static threshold curve is defined by a pair of threshold height and threshold time position. The IC supports in total 13 threshold points (first fix and remaining adjustable). All values are stored in non-volatile memory and have to be transmitted after power-up.

Threshold height

The device supports setting the height of the thresholds with a resolution of 5 counts in the given range with an offset of 4. The height of threshold points (THVAL0 and THVAL1) is fixed to maximum amplitude during SEND mode and to THVAL2 during RECEIVE mode.

Threshold position

The absolute time position of each threshold point is fix according to the settings of T_{CORR} and T_{SHIFT} , except the threshold value at THPOS1. This position is correlated to the actual ringing time, if NFTG is active.

Threshold value interpolation

The threshold values are connected to the threshold curve as follows:

- The threshold height from measurement start (at THPOS0) to the first threshold point (at THPOS1) is kept constant with a maximum amplitude.
- From THPOS1 to THPOS2 the device uses an exponential interpolation algorithm to reduce the threshold height to a lower value with a similar shape as the envelope signal. The exponential interpolation has a fix time constant, which might result in not reaching THVAL2 at THPOS2. The linear interpolation then starts with the value of the exponential interpolation at THPOS2.
- Between the threshold points THPOS2 and THPOS13, the IC uses a linear interpolation between two adjacent threshold points.
- After the last threshold point at THPOS13 the threshold curve is kept constant with the value defined by THVAL13.
- If the measurement range ends before the last threshold point, the last interval is still interpolated between the last threshold point inside and the first threshold point outside the measurement range.

Receive Mode

- In receive mode the first threshold interval is constant between THPOS0 and THPOS2 with THVAL2.

Table 5.6.3.1-1: Threshold curve

THPOS	Threshold height [LSB]	Interpolation type to next position	Time of THPOS [ms]
Start of measurement	SEND: 31 (fix) RECEIVE: THVAL2	constant	0
1	SEND: THVAL1 = 31 (fix) RECEIVE: THVAL2	exponential	$1.25 + T_{\text{CORR}} + T_{\text{SHIFT}}$
2	THVAL2	linear	$1.5 + T_{\text{CORR}} + T_{\text{SHIFT}}$
3	THVAL3	linear	$2 + T_{\text{CORR}} + T_{\text{SHIFT}}$
4	THVAL4	linear	$2.5 + T_{\text{CORR}} + T_{\text{SHIFT}}$
5	THVAL5	linear	$3 + T_{\text{CORR}} + T_{\text{SHIFT}}$
6	THVAL6	linear	$3.5 + T_{\text{CORR}} + T_{\text{SHIFT}}$
7	THVAL7	linear	$4 + T_{\text{CORR}} + T_{\text{SHIFT}}$
8	THVAL8	linear	$4.5 + T_{\text{CORR}} + T_{\text{SHIFT}}$
9	THVAL9	linear	$5 + T_{\text{CORR}} + T_{\text{SHIFT}}$
10	THVAL10	linear	$7 + T_{\text{CORR}} + T_{\text{SHIFT}}$
11	THVAL11	linear	$9 + T_{\text{CORR}} + T_{\text{SHIFT}}$
12	THVAL12	linear	$13 + T_{\text{CORR}} + T_{\text{SHIFT}}$
13	THVAL13	constant	$17 + T_{\text{CORR}} + T_{\text{SHIFT}}$
End of measurement	-	-	variable

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The filter bandwidth in combination with the number of pulses influences the filter delay. To adapt the threshold position to this delay time, the correction time T_{CORR} is added to each THPOS time depending on the chosen configuration. This is shown in table Table 5.6.3.1-2.

Table 5.6.3.1-2: Correction time T_{CORR}

NPULSES	T_{CORR} for FILTER_CFG='0' [us]	T_{CORR} for FILTER_CFG='1' [us]
4	-190	-184
8	-104	-114
12	-52	-53
16	0	0
20	104	74
24	155	121
28	207	121
32	243	121

Remark: The correction time T_{CORR} is adjusted for transducers with a driving frequency of 58kHz at room temperature, but also fits to other transducer frequencies much better than no compensation.

An additional shift of all threshold positions can be used to adapt the short range threshold points to different transducer types. If THSFT_CFG > '00', the threshold position time shift is active and can be chosen regarding the used transducer type. All threshold positions THPOS1 to THPOS13 are delayed by T_{SHIFT} according to following table:

Table 5.6.3.1-3: Time shift (T_{SHIFT}) for static threshold positions

THSFT_CFG	T_{SHIFT} in [us]
0	0
1	200
2	400
3	600

5.6.3.2 Values of static threshold generation

For each profile the threshold curve between THVAL2 and THVAL13 can be scaled individually by THRESSCALE in SEND and RECEIVE mode. In RECEIVE mode the threshold curve can be additionally scaled for all measurement profiles with THRESHSCALE_REC.

The scaling can be used to adjust the echo height regarding to the choice of NPULSES and FILTER_CFG.

SEND mode

In SEND mode the final threshold values for THVAL=thres1 ... THVAL=thres31 can be derived by:

maximum of [(THVAL * scaling_factor_{ProfileA}) and 4] = final_Threshold_{ProfileA}

maximum of [(THVAL * scaling_factor_{ProfileB}) and 4] = final_Threshold_{ProfileB}

maximum of [(THVAL * scaling_factor_{ProfileC}) and 4] = final_Threshold_{ProfileC}

Due to a scaling_factor < 1.0 a final threshold value might be calculated to a value below 4. These values will be set to 4. The scaling_factor can be configured by THRESSCALE individually for each profile.

Exception: THVAL=thres0

A static threshold of 0 is possible for THVAL=thres0 and can be used to definitely achieve a value below the automatic threshold generation.

Table 5.6.3.2-1: Scaling of Threshold

THRESSCALE[1]	THRESSCALE[0]	scaling_factor
0	0	0.5
0	1	0.75
1	0	1.0
1	1	1.25

Depending on the scaling setup the threshold values (THVAL) can be chosen from the following table. Each threshold height (THVAL) can take a value between thres0 and thres31. The final threshold value additionally depends on the scaling_factor.

Table 5.6.3.2-2: Threshold values resulting from scaling factor

Threshold height value	Threshold value with scaling 1.0	Threshold height value	Threshold value with scaling 1.0
thres0	0	thres16	42
thres1	5	thres17	46
thres2	6	thres18	51
thres3	8	thres19	56
thres4	9	thres20	61
thres5	11	thres21	67
thres6	13	thres22	73
thres7	15	thres23	79
thres8	17	thres24	86
thres9	19	thres25	94
thres10	22	thres26	102
thres11	25	thres27	111
thres12	28	thres28	121
thres13	31	thres29	131
thres14	34	thres30	142
thres15	38	thres31	155

Be careful: If THVAL = thres0 is chosen as threshold value and automatic threshold is off (ATG_CFG='00') or active only after the chosen position, then: THVAL < envelope signal (this indicates an echo).

RECEIVE mode

In 'RECEIVE' mode the threshold values for THVAL=thres1 ... THVAL=thres31 can be scaled by selection of THRESSCALE_REC parameter. They also depend on the chosen scaling in SEND mode and can be derived as follows:

maximum of [(THVAL * scaling_factor_rec * scaling_factor_{ProfileA}) and 4] = final_Threshold_{ProfileA}

maximum of [(THVAL * scaling_factor_rec * scaling_factor_{ProfileB}) and 4] = final_Threshold_{ProfileB}

maximum of [(THVAL * scaling_factor_rec * scaling_factor_{ProfileC}) and 4] = final_Threshold_{ProfileC}

Table 5.6.3.2-3 shows the scaling for the receive threshold scaling_factor_rec:

Table 5.6.3.2-3: Scaling between Receive and Send mode

THRESSCALE_REC[1]	THRESSCALE_REC[0]	scaling_factor_rec
0	0	0.5
0	1	0.625
1	0	0.8125
1	1	1

5.6.3.3 Automatic threshold generation (ATG)

The automatic threshold generation (ATG) can be activated by ATG_CFG.

An activation means a combination of static and dynamic threshold, which is depicted in Figure 5.6.3.3-1.

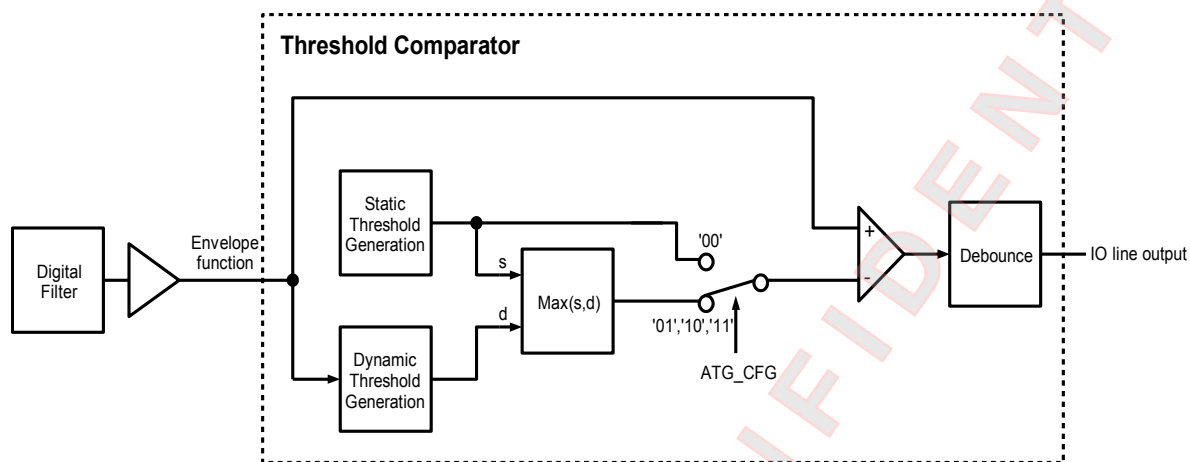


Figure 5.6.3.3-1: Threshold generation using ATG_CFG

The dynamic threshold is continuously calculated and depends on the time constant (ATG_TAU) and the multiplier (ATG_ALPHA). The maximum of the static threshold value and the dynamic calculated value is used as threshold.

The dynamic threshold parameters can be programmed with the 'THRES_SETUP' command within the volatile memory. See chapter 5.9.3 for more details.

The automatic threshold generation can be activated at different starting times with setting ATG_CFG:

Table 5.6.3.3-1: ATG_CFG

ATG_CFG[1]	ATG_CFG[0]	value
0	0	off
0	1	THPOS3 + T _{CORR}
1	0	THPOS7 + T _{CORR}
1	1	THPOS11 + T _{CORR}

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The time constant defines the fall time and can be set by ATG_TAU:

Table 5.6.3.3-2: ATG_TAU

ATG_TAU[1]	ATG_TAU[0]	speed
0	0	slow
0	1	medium
1	0	fast1
1	1	fast2

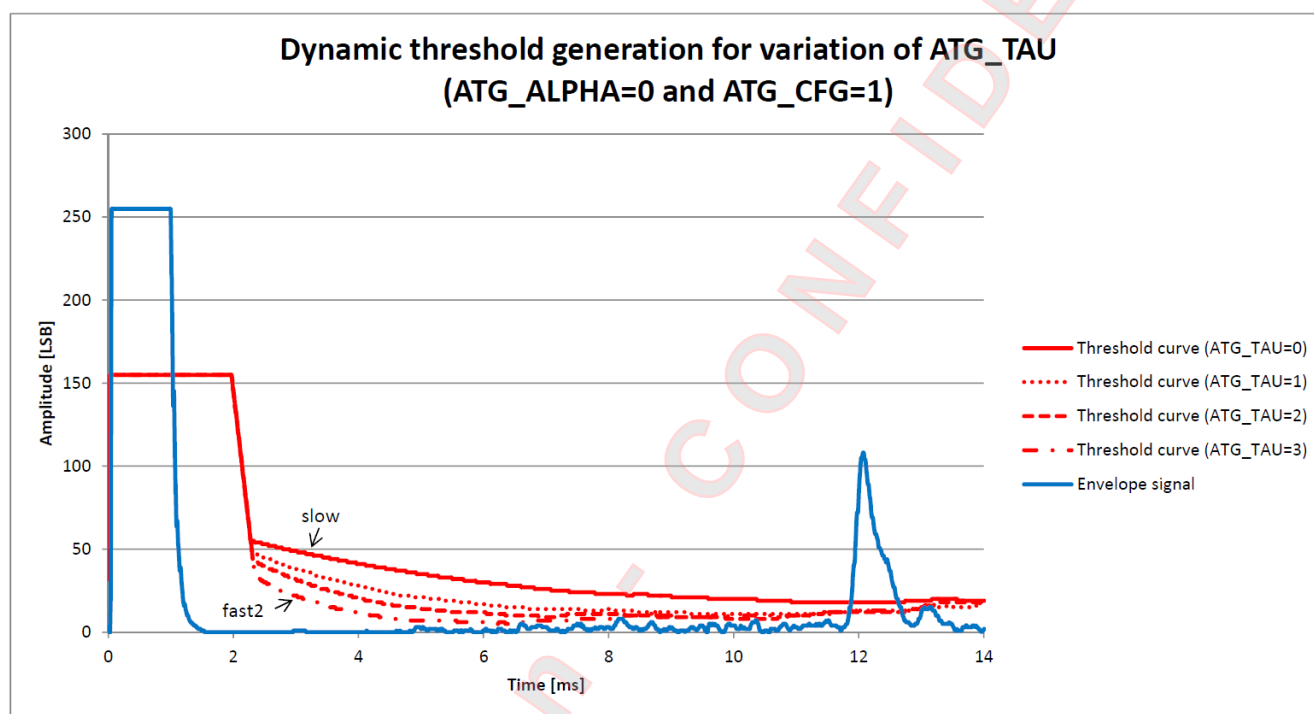


Figure 5.6.3.3-2: Dynamic threshold for ATG_TAU=varied, ATG_ALPHA=0, ATG_CFG=1

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The multiplier sets the signal-to-noise ratio between the envelope function and the used threshold.
It can be defined by ATG_ALPHA:

Table 5.6.3.3-3: ATG_ALPHA

ATG_ALPHA	value
0	sensitive (default)
1	less sensitive (optional)

The effect of the chosen ATG_ALPHA is depicted in following figure.

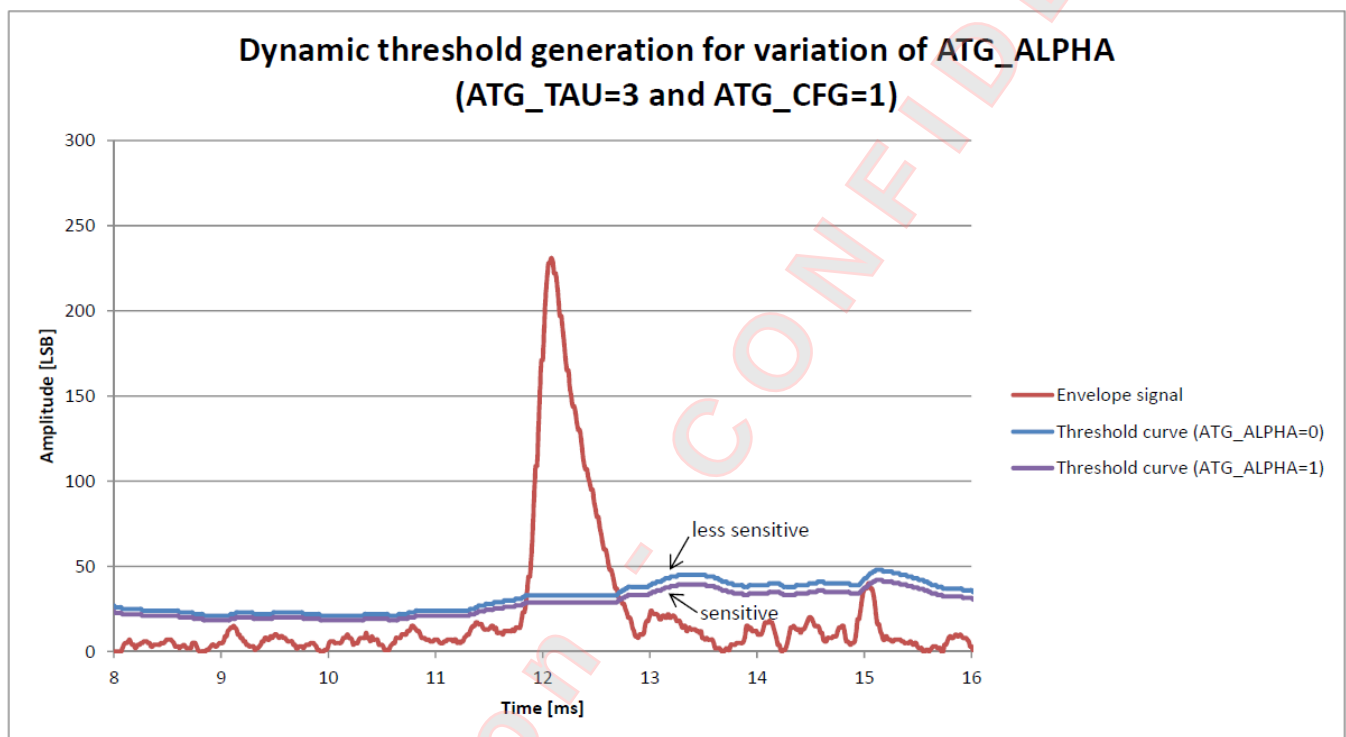


Figure 5.6.3.3-3: Dynamic threshold for ATG_TAU=3, ATG_ALPHA=varied and ATG_CFG=1

As soon as the envelope of the echo signal is above the dynamic threshold, the dynamic threshold stays constant.

5.6.3.4 Noise measurement during SEND and RECEIVE command

As increased ambient noise might disturb the measurement in SEND or RECEIVE mode, the IC measures the ambient super-sonic noise directly before entering SEND or RECEIVE mode. The ambient noise measurement is performed during the low phase of all 'SEND' or 'RECEIVE' commands. The output of the digital filter is compared to a threshold value which can be selected by bits NOISE_CFG according to Table 5.6.3.4-1. This noise measurement is performed with the amplification $G_{ANA} + G_{DIG} + G_{STC}$.

The result of the noise measurement is stored each 'SEND' or 'RECEIVE' command and can either be read out with the IO command 'READ_STATUS' or within the status information after each 'SEND' or 'RECEIVE' command.

Table 5.6.3.4-1: Threshold value for noise measurement depending of bits NOISE_CFG

NOISE_CFG[1]	NOISE_CFG[0]	Threshold value
0	0	16
0	1	32
1	0	64
1	1	128

5.6.3.5 Echo detection type

The IC offers two different types of echo detection:

- echo width detection
- echo peak detection (EPD)

In case of EPD='0' echo peak detection is inactive, which means: echo width detection is active. The IO-line is pulled down when the envelope signal crosses the threshold curve. The detected position of an obstacle depends on the ratio between the echo maximum and the crossing point of threshold curve with envelope signal.

In case of EPD='1' echo peak detection (EPD) is active. This means the IO-line is pulled down at the first maximum after the envelope signal crossed the threshold curve. The detected position of an obstacle nearly stays constant even with varied gain or threshold crossing.

The behavior of echo width and echo peak detection are depicted in the following figures. Two different obstacles were placed at the same distance to the sensor.

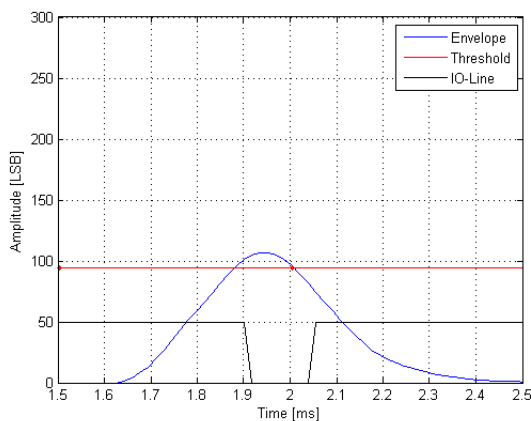


Figure 5.6.3.5-1: EPD='0' with low echo height

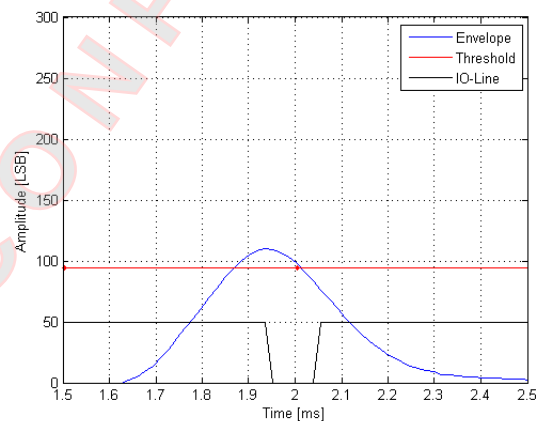


Figure 5.6.3.5-2: EPD='1' with low echo height

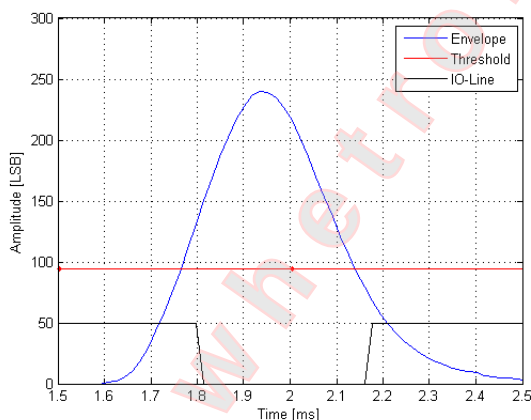


Figure 5.6.3.5-3: EPD='0' with high echo height

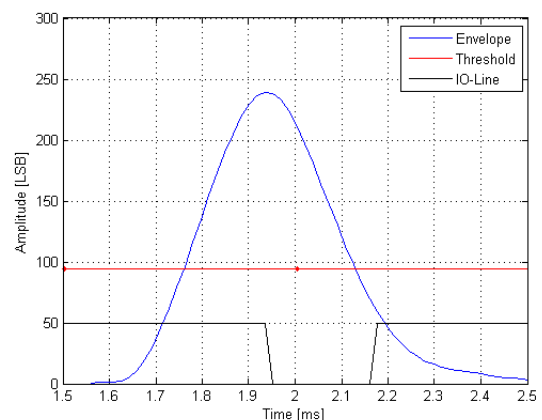


Figure 5.6.3.5-4: EPD='1' with high echo height

5.6.3.6 Echo peak detection during overdrive

Even if an echo signal overdrives within the 8 Bit range, the correct maximum position can be detected with EPD='1'. Echo height is calculated before digital gain and limitation is applied.

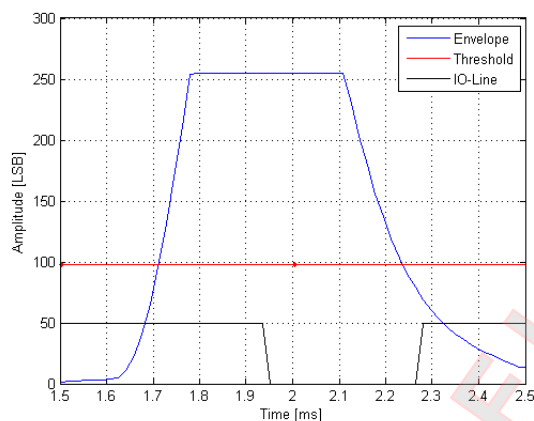


Figure 5.6.3.6-1: Echo peak detection with overdrive

5.6.4 Signal enhancement

5.6.4.1 Fast Time Constant (FTC)

The Fast Time Constant (FTC) method improves the echo detection by a differentiation algorithm, that is applied to the envelope function. It can be activated with the FTC bit, which is stored in the EEPROM memory.

If the FTC method is activated (FTC='1'):

- the FTC algorithm starts after the ringing phase
- the noise floor is reduced
- the envelope output is robuster against external disturbance

The following figures show the FTC behavior during a measurement, once with FTC active, once without. The measurement is done with the same environment containing two obstacles at different distances and angle to the transducer and additional noise at the end of the measurement.

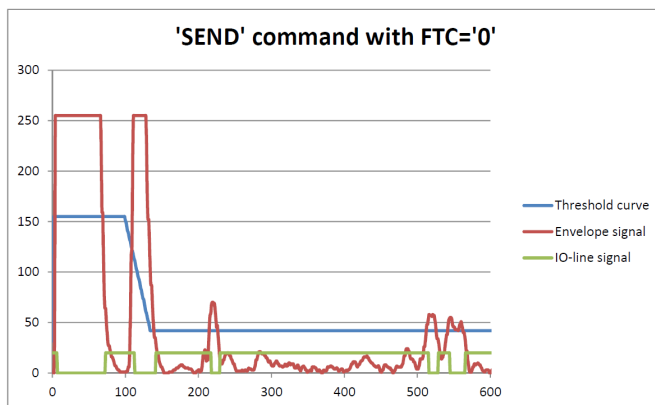


Figure 5.6.4.1-1: Measurement with FTC='0'

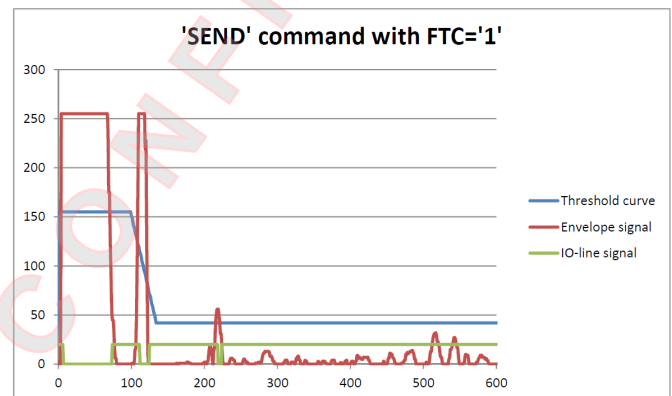


Figure 5.6.4.1-2: Measurement with FTC='1'

5.6.4.2 Near Field Threshold Generation (NFTG)

Near-field threshold generation results in a dynamical shift of the first threshold position depending on the ringing time of the transducer to optimize the near field detection especially for close obstacles besides the main beam. It can be activated with the NFTG bit, which is stored in the volatile memory.

- The position of THPOS1 is shifted towards THPOS2 until the envelope curve drops below the threshold.
- If THPOS1 is shifted too close toward THPOS2 the exponential interpolation may not reach the final value of THVAL2. In this case the linear interpolation between THVAL2 and THVAL3 starts with the value of the exponential interpolation at THPOS2.
- If THPOS1 reaches THPOS2, then THVAL2 = THVAL1. The shift of THPOS1 is limited to THPOS2.

The following figure shows the NFTG behavior of an obstacle besides the main beam and the threshold behavior with and without activated NFTG.

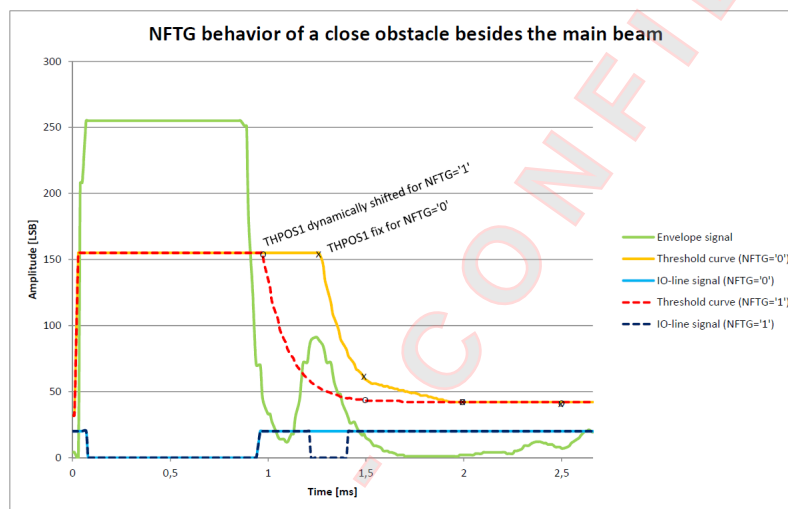


Figure 5.6.4.2-1: NFTG examples

5.6.4.3 Ringing time (RT)

The end of ringing time can be indicated by the envelope function after digital processing or by a comparator using the analog input signal. When the IO line behavior during ringing is controlled by a comparator (RT_CFG='1') the ringing time is independent from G_ANA, G_DIG and digital filter.

The configuration can be chosen with the RT_CFG bit, which is stored in the volatile memory.

The following figure shows the ringing time behavior of the IO-line:

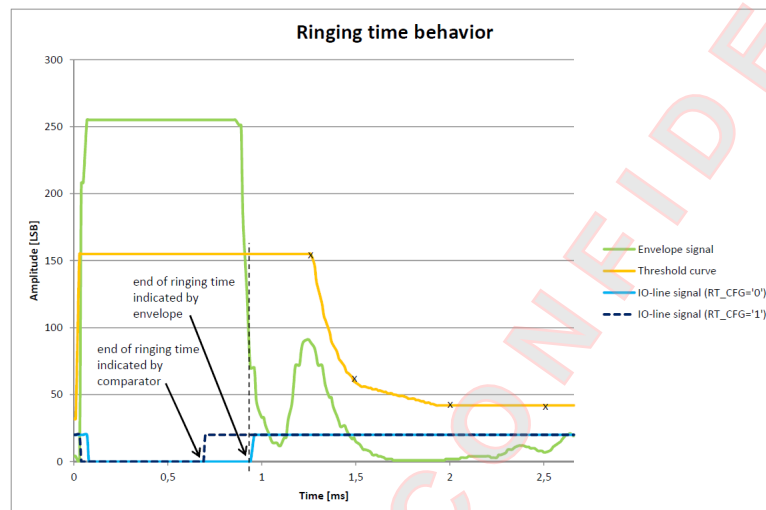


Figure 5.6.4.3-1: Ringing time behavior example

5.6.5 Measurement time

For each profile an individual measurement time can be set by TMEAS, which is stored in the measurement setup, using the 'MEAS_SETUP' command.

Table 5.6.5-1: TMEAS

TMEAS[2]	TMEAS[1]	TMEAS[0]	Measurement time [ms]	Corresponding length[m]
0	0	0	$8.75 + T_{\text{CORR}}$	1.5
0	0	1	$11.66 + T_{\text{CORR}}$	2
0	1	0	$14.58 + T_{\text{CORR}}$	2.5
0	1	1	$17.49 + T_{\text{CORR}}$	3
1	0	0	$23.32 + T_{\text{CORR}}$	4
1	0	1	$29.15 + T_{\text{CORR}}$	5
1	1	0	$34.98 + T_{\text{CORR}}$	6
1	1	1	$46.65 + T_{\text{CORR}}$	8

Remark: Distance information is calculated with an ultrasound speed of 343m/s.

5.7 Temperature sensor

The integrated temperature sensor may be used for an adjustment of transducer frequency or to monitor the temperature of IC.

- The 'READ_TEMP' command return the current temperature value.
- The first valid value after power-on or wake-up is available after $TSENSE_{T_DEL}$.
- The detected temperature is a junction temperature of the integrated circuit and not of the IC ambient.

The command 'READ_TEMP' returns a 10 bit value corresponding to the IC temperature. A higher ADC value $TSENS_C$ represents a higher temperature T. To calculate the temperature from the read value $TSENS_C$ the following formula can be used:

$$T(TSENS_C) = 25^{\circ}C + (TSENS_C - TSENS_{C_RT}) \cdot TSENS_{SLOPE}$$

To get an exact temperature, use the measured $TSENS_{C_RT}$ value at 25°C.

5.8 Memory and device configuration

The E524.08/09 supports two kinds of memory:

- Registers (volatile memory)
- EEPROM cells (non-volatile memory)

The 3 registers "Measurement Setup", "Threshold Setup" and "Calibration Setup" are used to configure the device.

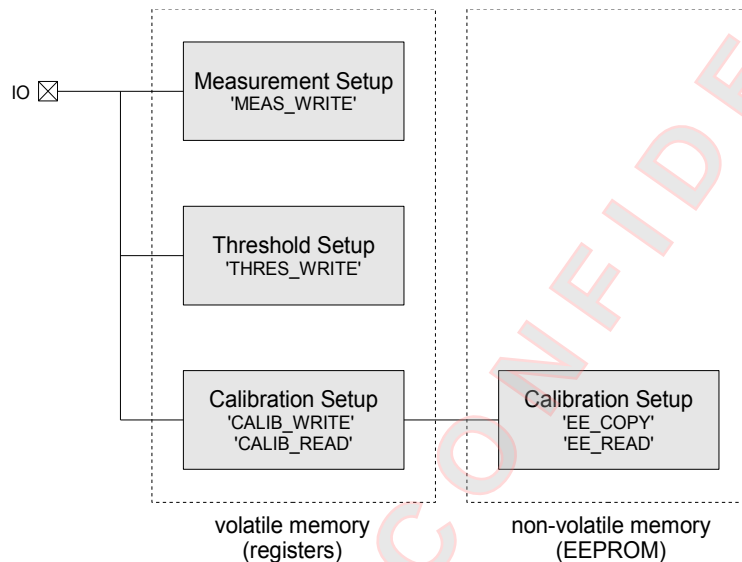


Figure 5.8-1: Memory and configuration overview

5.8.1 "Measurement Setup" and "Threshold Setup"

The registers "Measurement Setup" and "Threshold Setup" contain the settings for different measurement setups and parameters for the echo detection. Both are set to their default values after power up and can be changed by sending new data by 'MEAS_SETUP' and 'THRES_SETUP' commands via the IO line. The communication protocols are secured with parity bits and a successful data transmission can be verified with the 'READ_STATUS' command. Due to the fact that the settings depend on the complete (parking) application, they are normally known and written by the master ECU.

5.8.2 "Calibration Setup"

Due to tolerances of the transducer membrane, the sound pressure level and the sensitivity of transducer the behavior of a single ultrasonic sensor may differ from other. Thus, the "Calibration Setup" contains values like transducer frequency, transmitting power of the driver stage and amplifier gain and must be adjusted by a customer end-of-line calibration to allow the same performance over all ultrasonic sensor modules.

After power up the register "Measurement Setup" is loaded from the EEPROM cells. It can be changed by the 'CALIB_WRITE' command and verified by the 'CALIB_READ' command. If the data should be stored permanently, the values in the register can be copied to the EEPROM cells by performing the 'EE_COPY' command. In order to verify a successful EEPROM programming the data can be read back by the 'EE_READ' command.

Remark: If the 'EE_READ' command is executed, the EEPROM data is first copied to the register and then transmitted on the IO line. In this case the register will be overwritten and may change the configuration of the IC.

5.9 IO Interface

The IO interface is used to communicate between the E524.08 or E524.09 and the hosting ECU. The most common 'SEND_A' and 'RECEIVE_A' commands are initiated by an ECU pulling pin IO low for T_{SND} or T_{REC} . For the additionally available 'SEND_B', 'SEND_C' and 'RECEIVE_B', 'RECEIVE_C' commands the ECU pulls IO low for T_{MEAS} and issues a 2 bit command sequence. For all other commands, the ECU pulls IO low for T_{CMD} and issues a 4 bit command sequence. A logic '0' is signaled with pulling the line low for T_{BIT0} and a logic '1' with T_{BIT1} as shown in Figure 5.9-1. The bit length and all other timing parameters are temperature dependent and depend on f_{OSC} .

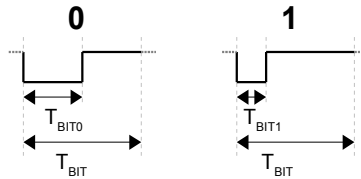


Figure 5.9-1: IO Pin bit encoding

E524.08

Data is sent between the E524.08 and ECU by modulating voltage on the supply wire VSUP_DATA. The application diagrams in Figure 7.1-1 and Table 7.2-1 show the location and size of the external components to ensure secure bidirectional communication. During communication, tank capacitors C_{SUP3} supplies the E524.08 at pin VSUP.

To send data to the E524.08, the supply line voltage has to be modulated by the ECU. A voltage at the pin IO below $V_{\text{IO_IL}}$ will be detected as a low level, a voltage above $V_{\text{IO_IH}}$ as a high level. A debouncing circuit suppresses interference shorter than T_{DEB} .

Data is transmitted from the E524.08 to the ECU by modulating pull-down current $I_{\text{PD_IO}}$ into pin IO. $I_{\text{PD_IO}}$ increases the supply current coming from the ECU via VSUP_DATA. Increasing $I_{\text{PD_IO}}$ will increase the voltage drop over R_i and decrease the voltage at $V_{\text{I_ECU}}$ which is recognized as a low level by the ECU. Consider plug and wire resistances for VSUP_DATA and GND connecting ultrasonic module and ECU to establish proper voltage levels at pin IO.

E524.09

Data is sent between the E524.09 and ECU by pulling the single line DATA low for defined periods of time. The application diagram in Figure 7.3-1 and Table 7.4-1 show the location and size of the external components to ensure secure communication via DATA.

Incoming data to the E524.09 is recognized as a low if V_{IO} is below $V_{\text{IO_IL}}$ of the voltage at VSUP. A debouncing circuit suppresses interference shorter than T_{DEB} . Outgoing data transmission to the ECU is accomplished by pulling pin IO low overcoming pull-up resistor $R_{\text{IO_PU_ECU}}$. To reduce EMI, the slew rate pulling low is limited to SR_{VIO} . The slew rate of the rising edge is defined by external circuitry.

5.9.1 Exchange of commands

The E524.08 and E524.09 support following 'SEND' and 'RECEIVE' commands from the hosting ECU.

Table 5.9.1-1: 'SEND' and 'RECEIVE' command structure summary

Command	Coding	Command Description
SEND_A	T _{SND}	Initiates a burst signal via the internal drivers followed by a measurement sequence and a status sequence. Measurement parameters of profile A from MEAS_SETUP register are used.
RECEIVE_A	T _{REC}	Initiates a measurement sequence without preceding burst, useful for triangulation in systems with multiple transducers. The measurement sequence is followed by a status sequence. Measurement parameters of profile A from MEAS_SETUP register are used.
SEND_B	T _{MEAS} + 10	Initiates a burst signal via the internal drivers followed by a measurement sequence and a status sequence. Measurement parameters of profile B from MEAS_SETUP register are used.
RECEIVE_B	T _{MEAS} + 00	Initiates a measurement sequence without preceding burst, useful for triangulation in systems with multiple transducers. The measurement sequence is followed by a status sequence. Measurement parameters of profile B from MEAS_SETUP register are used.
SEND_C	T _{MEAS} + 11	Initiates a burst signal via the internal drivers followed by a measurement sequence and a status sequence. Measurement parameters of profile C from MEAS_SETUP register are used.
RECEIVE_C	T _{MEAS} + 01	Initiates a measurement sequence without preceding burst, useful for triangulation in systems with multiple transducers. The measurement sequence is followed by a status sequence. Measurement parameters of profile C from MEAS_SETUP register are used.

The E524.08 and E524.09 support following commands from the hosting ECU besides the 'SEND' and 'RECEIVE' commands.

Table 5.9.1-2: Command structure summary

Command	Coding	Command Description
THRES_SETUP	T _{CMD} + 0000	Writes individual values for the static threshold generation and its scaling in receive mode into the device. Additionally the dynamic threshold generation can be activated and configured.
MEAS_SETUP	T _{CMD} + 0001	Writes individual measurement setup for up to 3 profiles (number of burst pulses, measurement time and scaling) into the device. Additionally settings for STC, noise and filter can be configured.
READ_STATUS	T _{CMD} + 0010	Verifies previous transmission of THRES_SETUP and MEAS_SETUP, reads current noise status, reads frequency deviation between driver and resonance frequency, reads overvoltage, undervoltage and reset state.
CAL_PULSES	T _{CMD} + 0011	Returns 2 pulses with a defined interval of TCAL for transducer driver frequency calibration purposes.
READ_TEMP	T _{CMD} + 0100	Returns the ADC value of the temperature sensor.
ENVELOPE_SEND_A	T _{CMD} + 0101	Initiates a burst signal via the internal drivers followed by a measurement sequence of profile A. Instead of the echo detection the envelope signal is shown on the IO line. (useful in 3-wire configuration)
ENVELOPE_REC_A	T _{CMD} + 0110	Initiates a measurement sequence of profile A without preceding burst. Instead of the echo detection the envelope signal is shown on the IO line. (useful in 3-wire configuration)
CALIB_WRITE	T _{CMD} + 1000	Allows to pre-load settings for the driver frequency, driver current, analog and digital amplifier gain into the device.
CALIB_READ	T _{CMD} + 1001	Reads the calibration registers to verify CALIB_WRITE.
EE_COPY	T _{CMD} + 1010	Copies the settings from the calibration register into the internal EEPROM.
EE_READ	T _{CMD} + 1011	Reads the EEPROM contents with three read voltage levels to verify successful programming.
READ_ID	T _{CMD} + 1100	A 24 bit chip ID is transferred on the IO line.
STANDBY	T _{CMD} + 1101	Initiates standby mode to decrease current consumption.
WAKE_UP	T _{CMD} + 1110	Initiates a wake up from standby mode.

5.9.2 'SEND' and 'RECEIVE' Commands

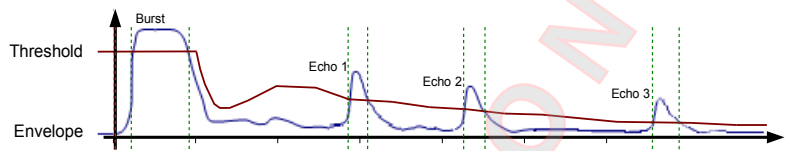
The IC can store 3 individual profiles, called profile A, profile B and profile C, which are used by the 3 SEND commands and the 3 RECEIVE commands. In each profile the number of pulses (NPULSES), the length of measurement time (TMEAS) and a threshold curve scaling (THRESHSCALE) can be stored in the volatile memory to configure the different profiles.

These profiles are intended to provide a short range and a long range measurement besides a normal range measurement. Alternatively they could be used to configure the threshold scaling and number of pulses for different temperatures.

Table 5.9.2-1: Available Profiles

Profile	NPULSES	TMEAS	THRESSCALE	'SEND' and 'RECEIVE' commands
Profile A	NPULSES_A	TMEAS_A	THRESSCALE_A	'SEND_A', 'RECEIVE_A'
Profile B	NPULSES_B	TMEAS_B	THRESSCALE_B	'SEND_B', 'RECEIVE_B'
Profile C	NPULSES_C	TMEAS_C	THRESSCALE_C	'SEND_C', 'RECEIVE_C'

Typical measurement:



Command examples:

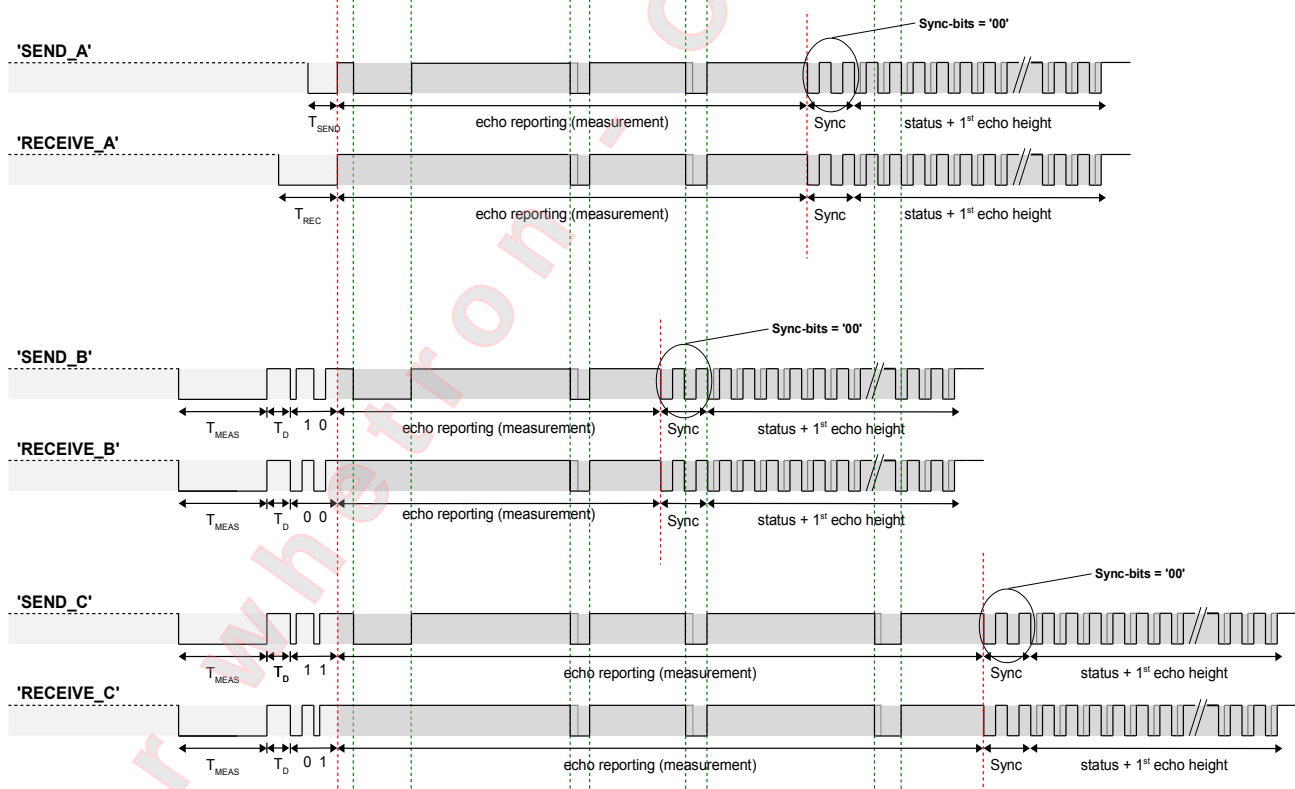


Figure 5.9.2-1: 'SEND' and 'RECEIVE' commands for typical measurements

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5.9.2.1 'SEND_A' command

The 'SEND_A' command initiates a burst followed by a measurement sequence. The SEND mode is requested from the ECU by holding Pin IO low for T_{SND} and is entered with the rising edge of the corresponding command.

As long as the E524.08 and E524.09 operate in SEND mode, no instructions coming over the IO-Interface are accepted. The interface is ready to receive new instructions when a high level remains at pin IO for at least T_{DEB} to debounce after SEND mode has ended.

The E524.08 and E524.09 can diagnose the proper operation of the transducer. After the 'SEND_A' command, the transducer driver is activated and sends out a burst for the length T_{TX} . This also causes high signal levels at the receiving amplifier and changes the level on the IO-line.

E524.08

For the E524.08 Figure 5.9.2.1-1 illustrates the 'SEND_A' command timing with an example of correct operation.

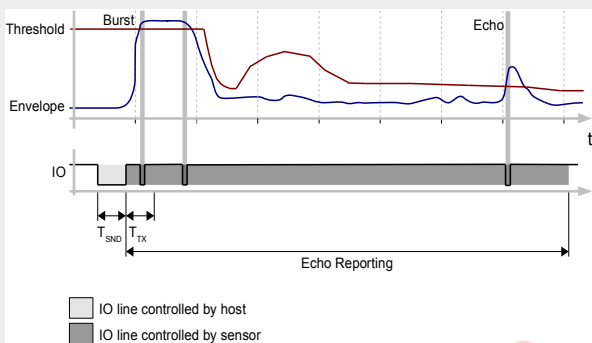


Figure 5.9.2.1-1: E524.08 'SEND_A' command timing

In the example above shows a typical envelope curve with one echo and a possible threshold curve. It shows the three different possibilities when the IO line is pulled down.

The first low pulse is reported during burst, the second low pulse indicates the end of ringing time and can be used to verify correct operation and the third pulse in this example is triggered due to a detected echo.

E524.09

For the E524.09, the duration of the low phase after T_{TX} can be used for diagnosis purposes. Figure 5.9.2.1-2 illustrates the 'SEND_A' command timing for the E524.09 with an example of correct operation.

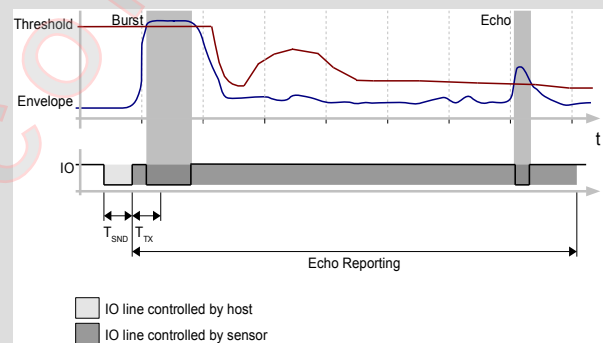


Figure 5.9.2.1-2: E524.09 'SEND_A' command timing

In the example above shows a typical envelope curve with one echo and a possible threshold curve. An echo is detected, when the signal line crosses the threshold line, which means that the comparator is triggered.

E524.08

The figure below shows that a crossing between threshold line and envelope will only be recognized on the IO_line if a debounce time of T_{DEB_ECHO} is maintained.

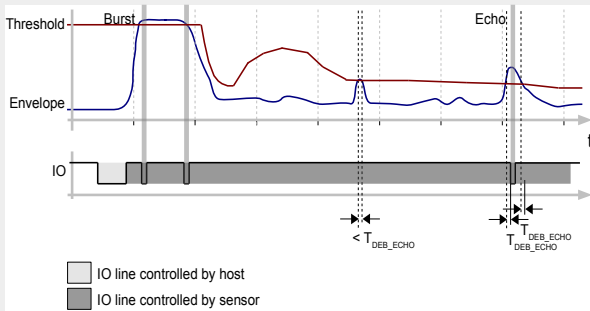


Figure 1: E524.08 'SEND_A' command timing with T_{DEB_ECHO}

E524.09

The figure below shows that a crossing between threshold line and envelope will only be recognized on the IO_line if a debounce time of T_{DEB_ECHO} is maintained.

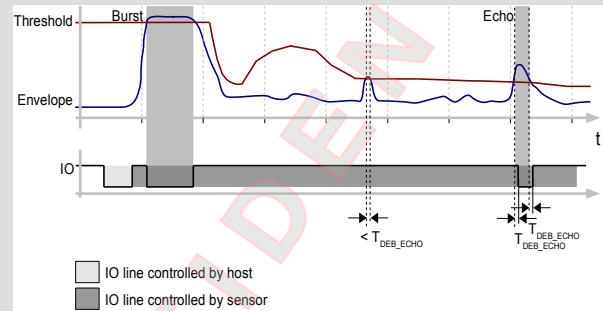


Figure 2: E524.09 'SEND_A' command timing with T_{DEB_ECHO}

5.9.2.2 'SEND_B' and 'SEND_C' command

'SEND_B' and 'SEND_C' commands are similar to 'SEND_A' command except:

1. The profile setups (NPULSES, TMEAS, THRESSCALE) are individually set using the 'MEAS_SETUP' command.
2. Each command has a different coding.

Please refer to Figure 5.9.2-1 for an overview and chapter 5.9.2.1 for detailed information.

5.9.2.3 'RECEIVE_A' command

A 'RECEIVE_A' command is identical to the 'SEND_A' command but skips the burst. The RECEIVE mode is requested from the ECU by holding Pin I/O low for T_{REC} and is entered with the rising edge of the corresponding command. A filter output signal exceeding the comparator threshold indicates a received echo. In RECEIVE mode the threshold sensitivity can be increased by a scaling factor set by $THRESSCALE_REC$.

As long as the E524.08 and E524.09 operate in RECEIVE mode, no instructions coming over the IO-Interface are accepted. The interface is ready to receive new instructions when a high level remains at pin IO for at least T_{DEB} to debounce after RECEIVE mode has ended.

E524.08

Figure 5.9.2.3-1 shows the timing for a E524.08 'RECEIVE_A' command.

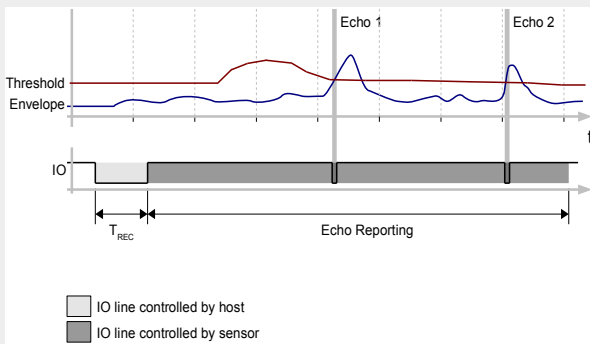


Figure 5.9.2.3-1: E524.08 'RECEIVE_A' command timing

The E524.08 operates purely in receiving mode for the entire sequence. In the example, two echoes are received and pull pin IO low for $T_{ECHO2WIRE}$. The thresholds are scalable with the selected scaling factor $THRESSCALE_REC$, part of the 'THRES_SETUP' command.

E524.09

Figure 5.9.2.3-2 shows the timing for a E524.09 'RECEIVE_A' command.

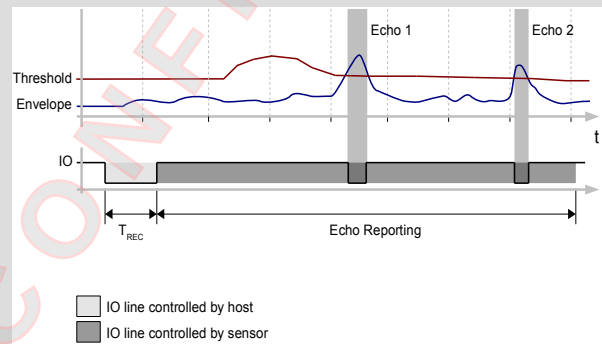


Figure 5.9.2.3-2: E524.09 'RECEIVE_A' command timing

The E524.09 operates purely in receiving mode for the entire sequence. In the example, two echoes are received and pull pin IO low. The thresholds are scalable with the selected scaling factor $THRESSCALE_REC$, part of the 'THRES_SETUP' command.

5.9.2.4 'RECEIVE_B' and 'RECEIVE_C' command

'RECEIVE_B' and 'RECEIVE_C' commands are similar to 'RECEIVE_A' command except:

1. The profile setups (NPULSES, TMEAS, THRESSCALE) are individually set using the 'MEAS_SETUP' command.
2. Each command has a different coding.

Please refer to Figure 5.9.2-1 for an overview and chapter 5.9.2.3 for detailed information.

5.9.2.5 Status information after 'SEND' and 'RECEIVE' commands

Each measurement (SEND or RECEIVE) is followed by two synchronisation-bits '00' and some status information containing the height of the first detected echo and flags to verify the noise level, hardware functionality and the ringing frequency deviation (freq_dev). This status information is intended to provide an overview of the status without sending additional commands. It can be used as a first diagnosis if the system is working properly.

The meaning of these bits is as follows:

Table 5.9.2.5-1: Status wrap up bits

Bit	Meaning
9	'0': No noise detected during last measurement command. '1': Noise detected during last measurement command.
8	'0': Last threshold value and measurement profile reception successful, no undervoltage, no overvoltage. '1': Indication that a supply voltage outside spec occurred or default threshold values or default measurement profiles are active.
7:6	'00': $ \text{freq_dev} \leq 3.12\%$ '01': $ \text{freq_dev} \leq 6.24\%$ '10': $ \text{freq_dev} \leq 9.36\%$ '11': $ \text{freq_dev} > 9.36\%$ or invalid measurement (default in RECEIVE mode)
5:0	6 Bit value of the echo height of the first detected echo. Upper 6 Bit of 8 Bit value are used. The returned value is the maximum of the region when the signal line is above the threshold. No echo during the measurement cycle results in '000000'.

More detailed status information is available by using the 'READ_STATUS' command.

5.9.3 'THRES_SETUP' command

The 'THRES_SETUP' command writes the threshold values of the threshold curve, threshold shift configuration, several control bits for automatic threshold generation and threshold scaling into the device. The threshold setup structure is depicted in the following table:

Table 5.9.3-1: Configuration of threshold setup

Bit	No. of bits	Function	Name	Default (dec)
68:64	5	2nd threshold value	THVAL2	10
63:59	5	3rd threshold value	THVAL3	10
58:55	5	4th threshold value	THVAL4	10
54:49	5	5th threshold value	THVAL5	15
48:44	5	6th threshold value	THVAL6	15
43:39	5	7th threshold value	THVAL7	15
38:35	5	8th threshold value	THVAL8	10
34:29	5	9th threshold value	THVAL9	10
28:25	5	10th threshold value	THVAL10	10
24:19	5	11th threshold value	THVAL11	0
18:14	5	12th threshold value	THVAL12	0
13:9	5	13th threshold value	THVAL13	0
8:7	2	Threshold position shift configuration: '00': 0us '01': 200us '10': 400us '11': 600us	THSFT_CFG	0
6:5	2	Automatic threshold configuration: '00': ATG off '01': ATG on after THPOS3 '10': ATG on after THPOS7 '11': ATG on after THPOS11	ATG_CFG	2
4:3	2	Time constant of ATG: '00': slow '01': medium '10': fast 1 '11': fast 2	ATG_TAU	1
2	1	Multiplier of ATG: '0': sensitive '1': less sensitive	ATG_ALPHA	0
1:0	2	Down scaling in receive mode: '00': 0.5 '01': 0.625 '10': 0.8125 '11': 1	THRESSCALE_REC	1

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The programming is secured with a blockwise calculation of even parity bits, which is depicted in the following diagram.

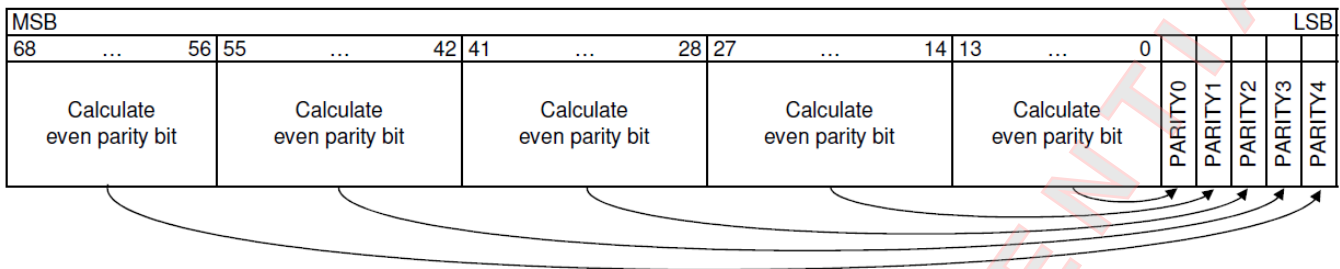


Figure 5.9.3-1: Blockwise parity calculation

The first bit transmitted on the IO line is the MSB of THVAL2 and the last bit is the parity bit PARITY4 depicted in Figure 5.9.3-2.

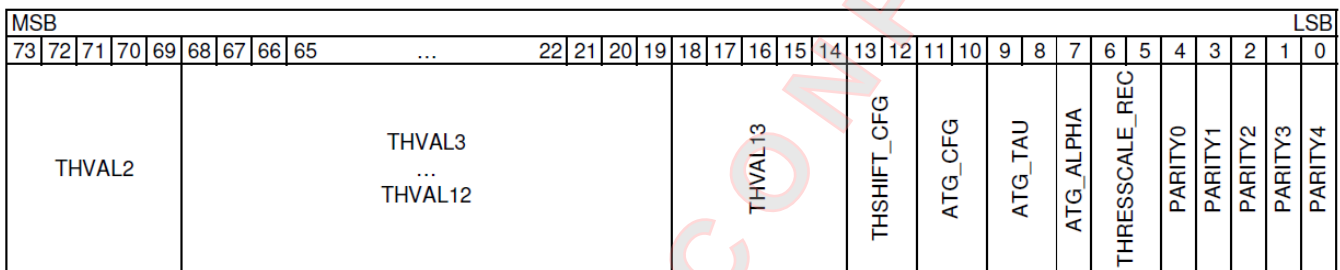


Figure 5.9.3-2: Transmitted data with 'THRES_SETUP' command

The 'THRES_SETUP' command timing diagram is shown in Figure 5.9.3-3.

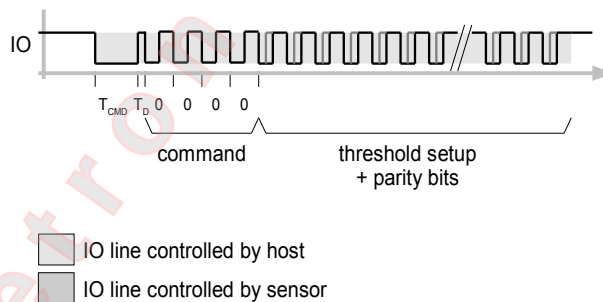


Figure 5.9.3-3: 'THRES_SETUP' command

The successful programming can be verified with the 'READ_STATUS' command.

Remark: Threshold data is stored in a volatile memory. Therefore the ECU has to write the threshold data every start up. After power on the default values are active.

5.9.4 'MEAS_SETUP' command

The 'MEAS_SETUP' command writes the configuration for measurement profile, STC, EPD, Filter and Noise into the device. The measurement setup structure is depicted in the following table:

Table 5.9.4-1: Configuration of measurement setup

Bit	No. of bits	Function	Name	default (dec)
34:32	3	Number of burst pulses - Profile A '000': 4 '001': 8 '010': 12 '011': 16 '100': 20 '101': 24 '110': 28 '111': 32	NPULSES_A	3
31:29	3	Measurement time - Profile A '000': 8.75 '001': 11.66 '010': 14.58 '011': 17.49 '100': 23.32 '101': 29.15 '110': 34.98 '111': 46.65	TMEAS_A	2
28:27	2	Threshold scaling factor - Profile A '00': 0.5 '01': 0.75 '10': 1.0 '11': 1.25	THRESSCALE_A	2
26:24	3	Number of burst pulses - Profile B	NPULSES_B	1
23:21	3	Measurement time - Profile B	TMEAS_B	0
20:19	2	Threshold scaling factor - Profile B	THRESSCALE_B	2
18:16	3	Number of burst pulses - Profile C	NPULSES_C	5
15:13	3	Measurement time - Profile C	TMEAS_C	6
12:11	2	Threshold scaling factor - Profile C	THRESSCALE_C	2
10	1	Ring time configuration '0': via envelope '1': via comparator	RT_CFG	0
9	1	Near field threshold generation: '0': off '1': on	NFTG	1
8	1	Fast Time Constant method: '0': off '1': on	FTC	0

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Bit	No. of bits	Function	Name	default (dec)
7	1	Echo Peak Detection '0': off '1': on	EPD	1
6:5	2	STC configuration: '00': STC off '01': $G_{STC}=24\text{dB}$ '10': $G_{STC}=30\text{dB}$ '11': $G_{STC}=\text{max}$	STC_CFG	2
4:3	2	STC start time: '00': 1.11ms '01': 1.40ms '10': 1.69ms '11': 1.98ms	STC_START	0
2:1	2	Noise threshold for noise measurement: '00': 16 '01': 32 '10': 64 '11': 128	NOISE_CFG	2
0	1	Digital Filter configuration '0': default '1': wide	FILTER_CFG	0

The programming is secured with a blockwise calculation of even parity bits, which is depicted in the following diagram.

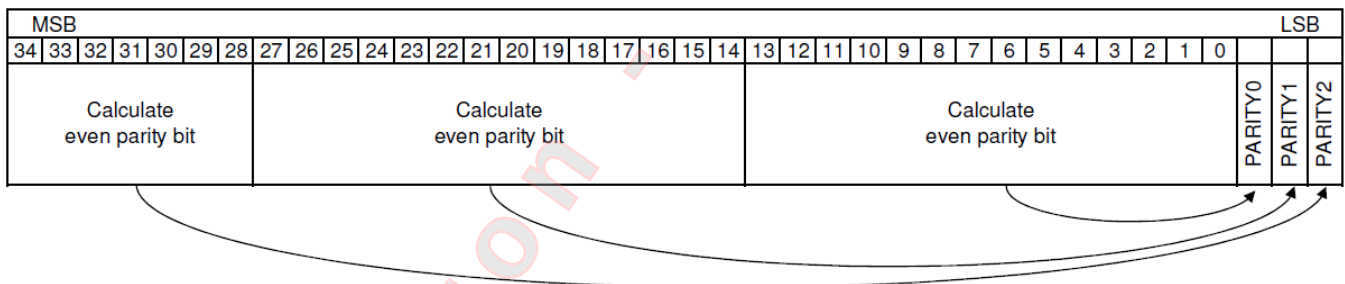


Figure 5.9.4-1: Blockwise parity calculation

The first bit transmitted on the IO line is the MSB of NPULSES_A and the last bit is the parity bit PARITY1:

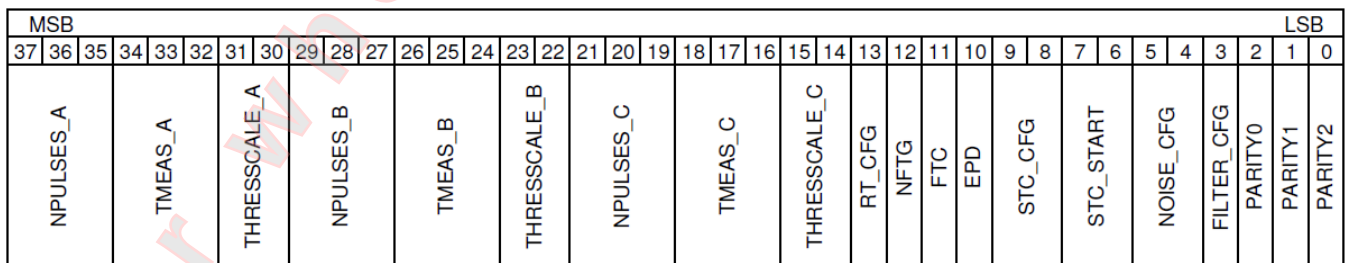


Figure 5.9.4-2: Transmitted data with 'MEAS_SETUP' command

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The 'MEAS_SETUP' command timing diagram is shown in Figure 5.9.4-3.

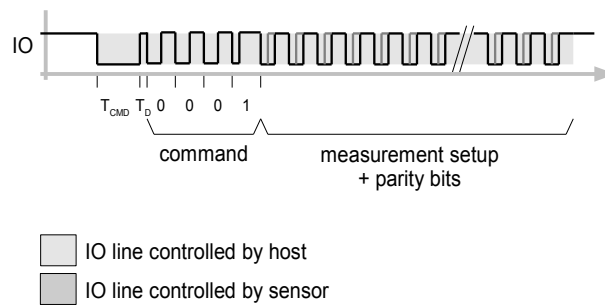


Figure 5.9.4-3: 'MEAS_SETUP' command

The successful programming can be verified with the 'READ_STATUS' command.

Remark: Measurement data is stored in a volatile memory. Therefore the ECU has to write the measurement data every start up. After power on the default values are active.

5.9.5 'READ_STATUS' command

The command 'READ_STATUS' can be used

- to verify the last transmission of the threshold setting configuration and the measurement setting configuration
- to get information about the frequency deviation between transducer driver and its resonance frequency and its validation
- to get information about undervoltage and overvoltage during the last measurement cycle

The noise level is analyzed during the low phase on the IO line of the send/receive request command.

The frequency deviation measurement of the latest 'SEND' is transmitted. The data of Bit0-5 should be evaluated before another 'SEND'. The meaning of these bits is as follows:

Table 5.9.5-1: Read Status: Bits 0-4

Bit 4:0	Meaning
01111	+11.7% frequency deviation between the transducer driver and its resonance frequency
01110	+10.92% frequency deviation between the transducer driver and its resonance frequency
...	
00001	+0.78% frequency deviation between the transducer driver and its resonance frequency
00000	0.0% frequency deviation between the transducer driver and its resonance frequency
11111	-0.78% frequency deviation between the transducer driver and its resonance frequency
...	
10001	-11.7% frequency deviation between the transducer driver and its resonance frequency
10000	-12.48% frequency deviation between the transducer driver and its resonance frequency

The frequency deviation between driver frequency and transducer resonance frequency is transmitted as a complement on two. Each digit equals 0.78% frequency deviation. The value is valid after the 'SEND' commands.

Table 5.9.5-2: Read Status: Bit 5

Bit 5	Meaning
0	Result of frequency measurement is invalid (default after RECEIVE mode)
1	Result of frequency measurement is valid

Table 5.9.5-3: Read Status: Bit 6

Bit 6	Meaning
0	Reset state or last threshold value reception according to Table 5.9.3-1 incorrect Default values active
1	Last threshold value reception according to Table 5.9.3-1 successful Transmitted threshold values active

Table 5.9.5-4: Read Status: Bit 7

Bit 7	Meaning
0	Reset state or last measurement profile reception according to Table 5.9.4-1 incorrect Default profiles active
1	Last measurement profile reception according to Table 5.9.4-1 successful Transmitted measurement profiles active

Table 5.9.5-5: Read Status: Bit 8

Bit 8	Meaning
0	No overvoltage
1	Overvoltage detected during last measurement cycle

Table 5.9.5-6: Read Status: Bit 9

Bit 8	Meaning
0	No undervoltage
1	Undervoltage detected during last measurement cycle

The 'READ_STATUS' command timing diagram is shown in Figure 5.9.5-1. The transmission starts with the MSB.

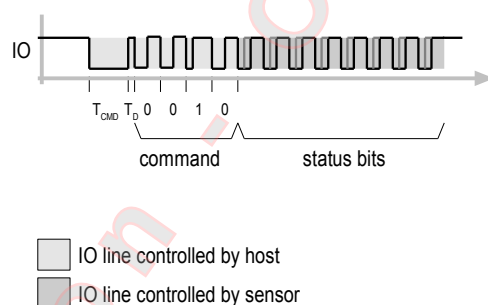


Figure 5.9.5-1: 'READ_STATUS' command

5.9.6 'CAL_PULSES' command

To calibrate the internal oscillator to the required frequency two pulses with a defined interval of T_{CAL} can be driven on the IO line. The interval starts and stops with a pulse of the width $T_{CAL,S}$. Due to the controlled slew rate, the ECU should observe the falling edge of the pulses. The following diagram shows the flow:

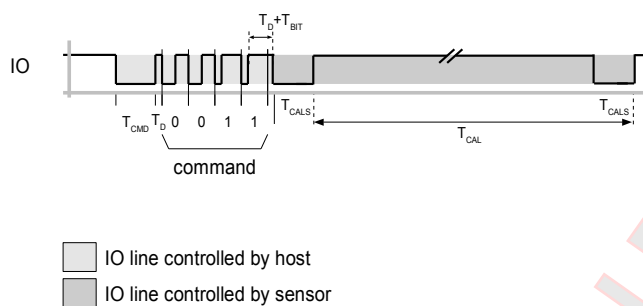


Figure 5.9.6-1: 'CAL_PULSES' command

5.9.7 'READ_TEMP'

The 'READ_TEMP' command timing diagram is shown in Figure 5.9.5-1. The transmission starts with the MSB.

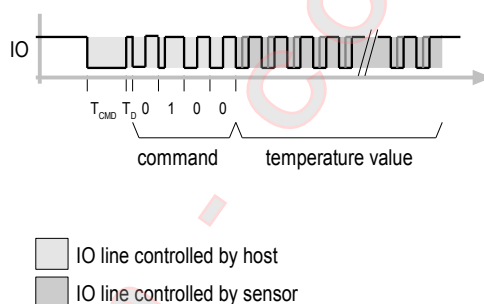


Figure 5.9.7-1: 'READ_TEMP' command

5.9.8 'ENVELOPE_SEND_A' and 'ENVELOPE_REC_A' commands

Instead of the echo detection in 'SEND' or 'RECEIVE' mode the envelope signal for 'SEND_A' or 'RECEIVE_A' (only profile A is possible) can be read out via IO line. This might be useful during evaluation or analysis of a moulded 3-wire system, where JTAG access is not possible.

The 'ENVELOPE_SEND_A' command initiates a burst signal via the internal drivers followed by a measurement sequence of profile A. Then the envelope signal is shown on the IO line.

The 'ENVELOPE_RECEIVE_A' command initiates a measurement sequence without preceding burst. After the measurement sequence of profile A the envelope signal is shown on the IO line.

The current output of the DAC with the external pull-up resistor results in a curve, that depicts the envelope function. The signal is internally inverted.

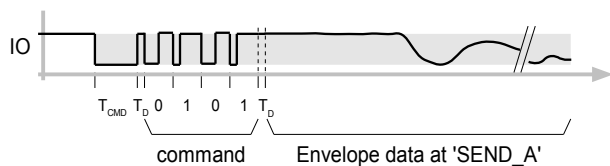


Figure 5.9.8-1: 'ENVELOPE_SEND_A' command

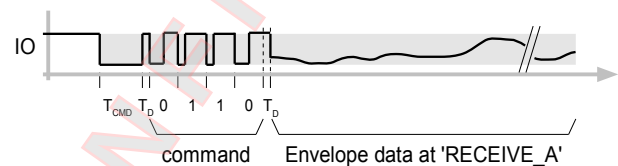


Figure 5.9.8-2: 'ENVELOPE_RECEIVE_A' command

5.9.9 EEPROM Programming

The internal 34 bit EEPROM holds values for driver frequency, driver current, digital and analog amplifier gain, oscillator trimming and free configurable customer bits. The programming and verification via the single IO pin is done in three steps.

'CALIB_WRITE' allows to pre-load initial settings. After calibration, 'EE_COPY' copies the final settings into the EEPROM. To verify successful programming, 'EE_READ' reads the register content back. To achieve a valid programming, 'CALIB_WRITE' must be sent before an 'EE_COPY'.

Both E524.08 and E524.09 are shipped with factory default settings. After assembly with a transducer, the optimum values for driver frequency, driver current, digital and analog amplifier gain are found in functional testing and calibration in the application of the customer. The EEPROM configuration is depicted in the following figure and table. The delivery state can be found within the electrical parameters of chapter 4.

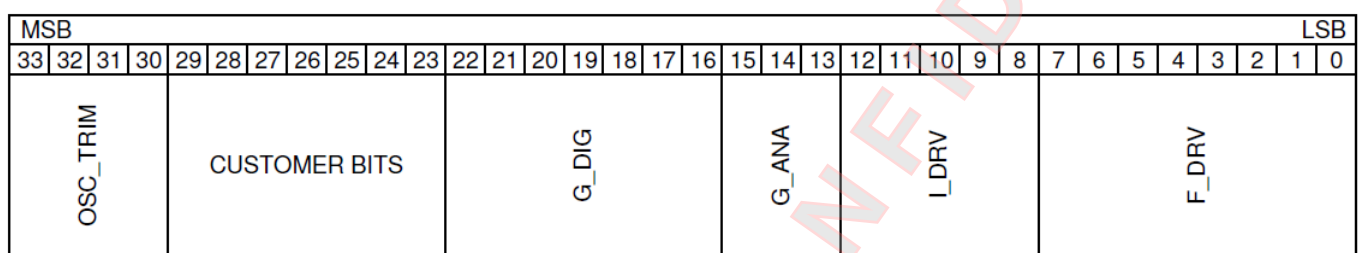


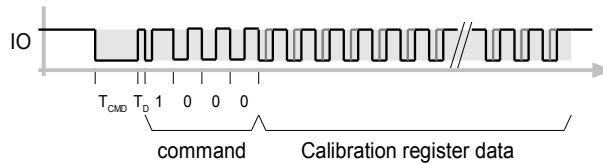
Figure 5.9.9-1: EEPROM organization

Table 5.9.9-1: EEPROM Bits

Bit	No. Bits	Function	Description
33:30	4	OSC_TRIM	oscillator trimming If no trimming is required use '0000'
29:23	7	CUSTOMER BITS	free configurable
22:16	7	G_DIG	Digital amplifier gain configuration
15:13	3	G_ANA	Analog amplifier gain adjustment
12:8	5	I_DRV	Transducer driver output current configuration
7:0	8	F_DRV	Driver frequency

5.9.9.1 'CALIB_WRITE' Command

As shown in Figure 5.9.9.1-1 in 'CALIB_WRITE', data bits follow directly after the command bits. The transmitted data bits overwrite previous register data, the new settings are valid for all functionality. The first bit transmitted on the IO line is bit 33 (MSB of OSC_TRIM) and the last bit is bit 0 (LSB of F_DRV).

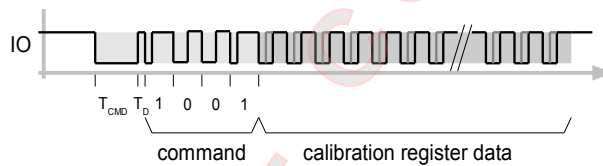


- ☐ IO line controlled by host
- ☐ IO line controlled by sensor

Figure 5.9.9.1-1: 'CALIB_WRITE' command

5.9.9.2 'CALIB_READ' command

To verify the data that has been written to the calibration data register the 'CALIB_READ' command can be used. It transfers the contents of the calibration data registers on the IO line.



- ☐ IO line controlled by host
- ☐ IO line controlled by sensor

Figure 5.9.9.2-1: 'CALIB_READ' command

5.9.9.3 'EE_COPY' Command

With the 'EE_COPY' command, the EEPROM cells are programmed with the current calibration register contents. After sending out the command, the voltage at the IO pin has to be raised to V_{PROG} within T_{VPROG} . As shown in Figure 5.9.9.3-1, the programming time at pin IO is T_{PROG} . Without a previous 'CALIB_WRITE' command the 'EE_COPY' doesn't lead to a change of the EEPROM content.

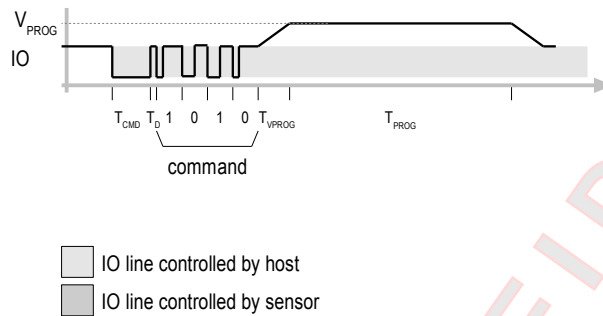


Figure 5.9.9.3-1: 'EE_COPY' command

5.9.9.4 'EE_READ' Command

To verify successful programming, the 'EE_READ' command loads the EEPROM data into calibration register and transmits the data at the IO pin, see Figure 5.9.9.4-1. The EEPROM data is read with a nominal, low and high read voltage and presented sequentially at the output IO indicating data integrity. Additionally, one VPROG_STATUS bit is appended to the end of the EEPROM bits being read each time. The VPROG_STATUS contains the output of a comparator observing the programming voltage during programming. A high level shows the programming voltage has been within limits, a low level indicates that the programming was not correct.

The first bit transmitted on the IO line is bit 33 (MSB of OSC_TRIM), the last bit is bit 0 (LSB of F_DRV) and an additional bit (VPROG_STATUS) with a nominal read voltage. The sequence is repeated with a low and a high read voltage. All in all 3 times (34+1) bit are transferred.

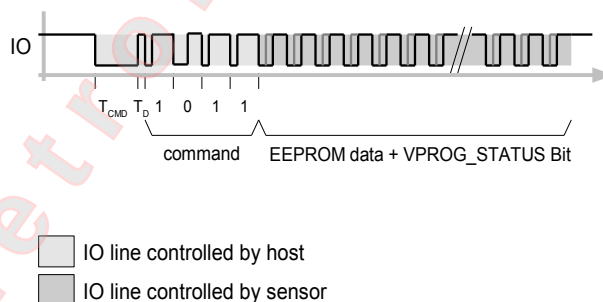


Figure 5.9.9.4-1: 'EE_READ' command

5.9.10 'READ_ID' command

A 24 bit chip ID is transferred on the IO line with the command 'READ_ID'. The MSB of the ID is transmitted first.

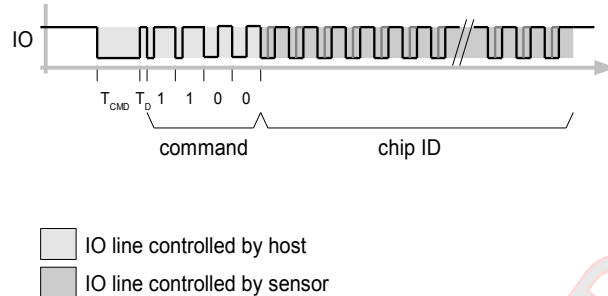


Figure 5.9.10-1: 'READ_ID' command

5.9.11 'STANDBY' and 'WAKE_UP' command

Besides normal operation E524.08 and E524.09 offer a current saving standby mode. All settings and status information keep their content.

- To enter standby mode send the 'STANDBY' command.
- To return to normal operation send the 'WAKE_UP' command.

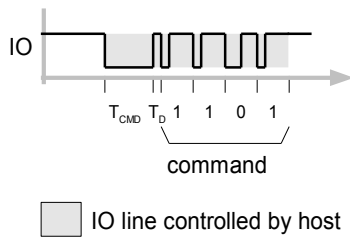


Figure 5.9.11-1: 'STANDBY' command

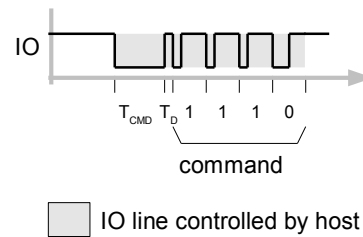


Figure 5.9.11-2: 'WAKE_UP' command

6 Package Reference

The IC is available in a Pb free, RoHs compliant, QFN20L4 plastic package. Package outline and dimensions are according to JEDEC MO-220 K, variant VGGD-5.

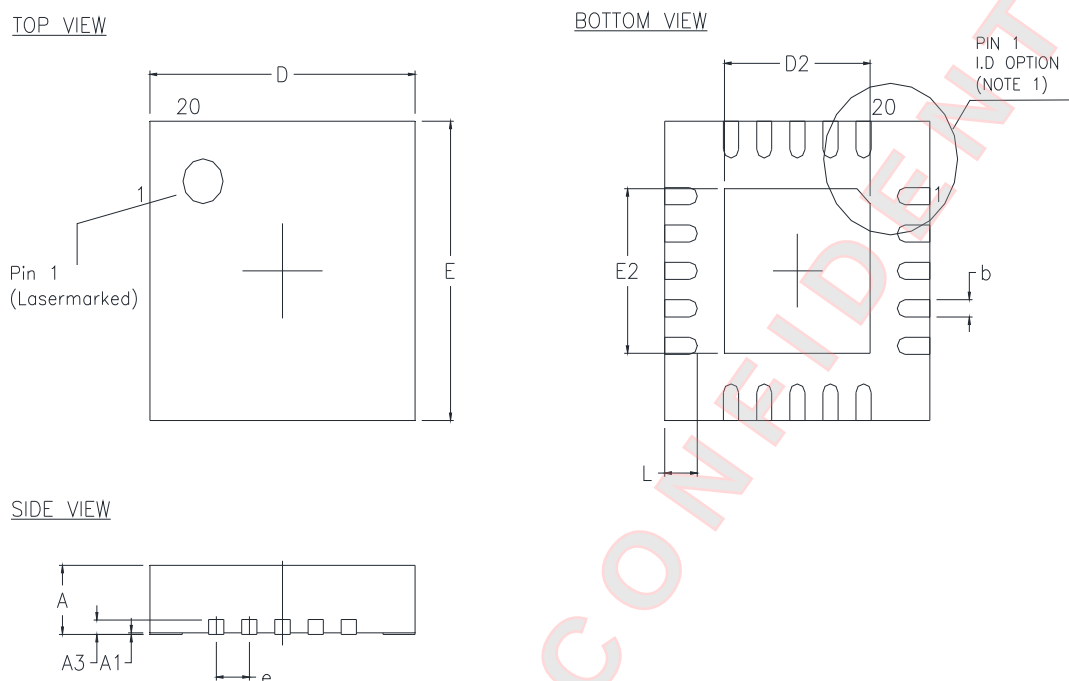


Figure 6-1: Package Outline

Note 1: Typical thermal resistance junction to ambient $R_{th,ja}$ is 45 °C/W, based on JEDEC standard JESD-51-2 (still air), JESD-51-5 (exposed pad soldered to PCB with thermal via's) and JESD-51-7 (4-layer PCB).

Note 2: Contact factory for specific location and type of pin 1 identification.

Table 6-1: Package Dimensions

Description	Symbol	mm			inch		
		min	typ	max	min	typ	max
Package height	A	0.80	0.90	1.00	0.031	0.035	0.039
Stand off	A1	0.00	0.02	0.05	0.000	0.00079	0.002
Thickness of terminal leads, including lead finish	A3	--	0.20 REF	--	--	0.0079 REF	--
Width of terminal leads	b	0.18	0.25	0.30	0.0071	0.0098	0.012
Package length / width	D / E	--	4.00 BSC	--	--	0.157 BSC	--
Length /width of exposed pad	D2 / E2	2.50	2.65	2.80	0.098	0.104	0.110
Lead pitch	e	--	0.5 BSC	--	--	0.020 BSC	--
Length of terminal for soldering to substrate	L	0.35	0.40	0.45	0.013	0.016	0.018
Number of terminal positions	N		20			20	

Note: The mm values are valid, the inch values contain rounding errors.

7 Typical Applications

7.1 Typical Application Circuit E524.08

The E524.08 supports a wide range of ultrasound transducers. Circuit diagrams may contain components not manufactured by EL MOS Semiconductor AG, which are included as means of illustrating a typical application. Elmos Semiconductor does not endorse or warrant performance specifications beyond the Electrical Characteristics for the E524.08. Please contact Elmos Semiconductor for further assistance and application notes.

Figure 7.1-1 and Table 7.2-1 show the typical operating circuit and external components for the E524.08. The ECU and transducer assembly are connected with two wires VSUP_DATA and GND. Data is exchanged via shared supply line VSUP_DATA.

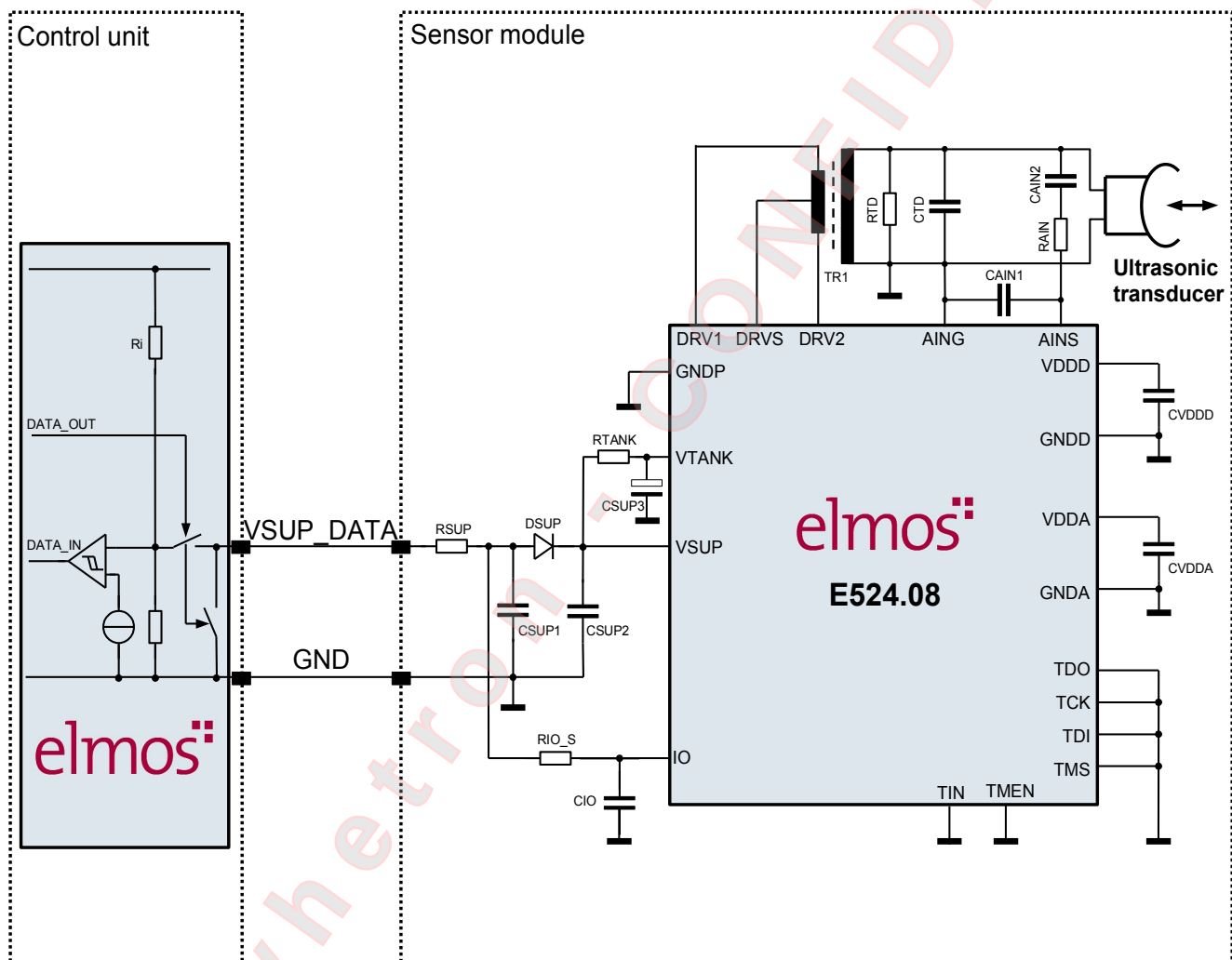


Figure 7.1-1: Application Circuit E524.08

7.2 External Components E524.08

Table 7.2-1: External component list E524.08

No.	Description	Condition	Symbol	Typ	Unit
1	Serial resistor in supply line	>330mW	RSUP ⁽¹⁾	15	Ω
2	Reverse polarity protection diode	BAS321 general purpose diode or equivalent	DSUP	>200	V _{Reverse}
3	Blocking capacitor for supply line		CSUP1	2.2	nF
4	Blocking capacitor for supply line		CSUP2 ⁽¹⁾	220	nF
5	Filter resistor for transducer tank	>330mW	RTANK ⁽¹⁾	82	Ω
6	Tank capacitor for supply line	Stores charge for burst, 50V, ESR<180mΩ	CSUP3 ⁽²⁾	68	uF
7	Serial resistor for IO line	5%, 1/8W	RIO_S	150	Ω
8	Capacitor for IO line	5%	CIO	4.7	nF
9	Filter capacitor for transducer signal	pay attention to voltage range	CAIN1	10	pF
10	Filter resistor for transducer signal	pay attention to power dissipation	RAIN	100	Ω
11	AC coupling capacitor for transducer signal	pay attention to voltage range	CAIN2	330	pF
12	Blocking capacitor for analog supply	10%, 50V, ESR<0.2Ω at 1MHz	CVDDA	100	nF
13	Blocking capacitor for digital supply	10%, 50V, ESR<0.2Ω at 1MHz	CVDDD	100	nF
14	Transducer resonance resistor	5%, >100V	RTD	adjust to transducer	
15	Transducer resonance capacitor	10%, >100V	CTD	adjust to transducer	
16	Transformer	Sugg. winding ration 1:1:8	TR1	3	mH
17	Transducer	e.g. Murata MA58MF14-7N	Ultrasonic transducer	58	kHz

⁽¹⁾ component values need to be adapted, if ISO Standard 7637-2 needs to be fulfilled not only on system level, but also on module level.

⁽²⁾ component value needs to be adapted to NPULSEES and IDRIV.

7.3 Typical Application Circuit E524.09

The E524.09 supports a wide range of ultrasound transducers. Circuit diagrams may contain components not manufactured by EL MOS Semiconductor AG, which are included as means of illustrating a typical application. Elmos Semiconductor does not endorse or warrant performance specifications beyond the Electrical Characteristics for the E524.09. Please contact Elmos Semiconductor for further assistance and application notes.

The typical operational circuit for the E524.09 is shown in Figure 7.3-1, Table 7.4-1 lists external components. Three wires (GND, VSUP and DATA) connect ECU and ultrasonic module.

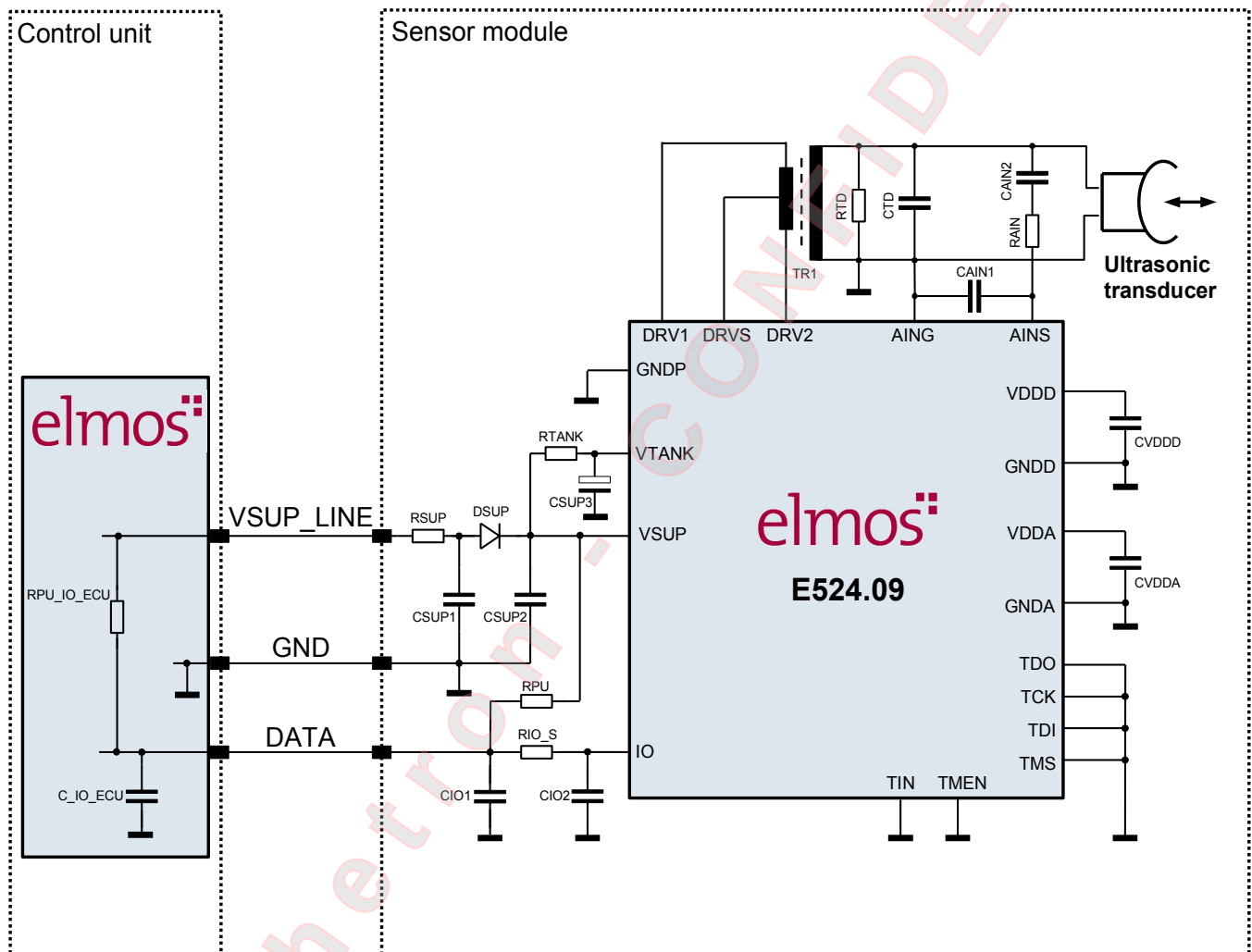


Figure 7.3-1: Application Circuit E524.09

7.4 External Components E524.09

Table 7.4-1: External component list E524.09

No.	Description	Condition	Symbol	Typ	Unit
1	Serial resistor in supply line	>330mW	RSUP ⁽¹⁾	15	Ω
2	Reverse polarity protection diode	BAS321 general purpose diode or equivalent	DSUP	>200	V _{Reverse}
3	Blocking capacitor for supply line		CSUP1	2.2	nF
4	Blocking capacitor for supply line		CSUP2 ⁽¹⁾	220	nF
5	Tank capacitor for supply line	Stores charge for burst, 50V, ESR<180mΩ	CSUP3 ⁽²⁾	68	uF
6	Filter resistor for transducer tank	>330mW	RTANK ⁽¹⁾	82	Ω
7	Serial resistor for IO line	5%, 1/8W	RIO_S	1	kΩ
8	Pull up resistor for IO line	5%, 1/8W	RPU ⁽³⁾	10	kΩ
9	Capacitor for IO line	5%	CIO1	330	pF
10	Capacitor for IO line	10%	CIO2	2.2	nF
11	Filter capacitor for transducer signal	pay attention to voltage range	CAIN1	10	pF
12	Filter resistor for transducer signal	pay attention to power dissipation	RAIN	100	Ω
13	AC coupling capacitor for transducer signal	pay attention to voltage range	CAIN2	330	pF
14	Blocking capacitor for analog supply	10%, 50V, ESR<0.2Ω at 1MHz	CVDDA	100	nF
15	Blocking capacitor for digital supply	10%, 50V, ESR<0.2Ω at 1MHz	CVDDD	100	nF
16	Transducer resonance resistor	5%, >100V	RTD	adjust to transducer	
17	Transducer resonance capacitor	5%, >100V	CTD	adjust to transducer	
18	Transformer	Sugg. winding ration 1:1:8	TR1	3	mH
19	Transducer	e.g. Murata MA58MF14-7N	Ultrasonic transducer	58	kHz

⁽¹⁾ component values need to be adapted, if ISO Standard 7637-2 needs to be fulfilled not only on system level, but also on module level.

⁽²⁾ component value needs to be adapted to NPULSES and IDRIV.

⁽³⁾ RPU must be chosen with respect to RPU_IO_ECU (e.g. 10kΩ // 10kΩ). The external elements should be chosen to set the rising edge and the falling edge of the communication via IO to the same slope.

8 Marking

8.1 Top Side

Table 8.1-1: Top Side

ELMOS (optional)
52408A or 52409A (optional)
XXXXU
YWW*#

Table 8.1-2: Marking of the Devices

<i>Signature</i>	<i>Explanation</i>
52408 or 52409	Elmos project number
A	Elmos project revision number
XXXX	Production lot number
U	Assembler Code
YWW	Year and week of assembly
*	Mask revision code
#	Elmos internal code

8.2 Bottom Side

No marking.

9 Reliability

9.1 EEPROM Data Retention

The data retention for the EEPROM within the given profile shown in Table 9.1-1 is statistically ensured according to AEC Q-100.

Table 9.1-1: EEPROM mission profile

<i>Junction Temperature in [°C]</i>	<i>Operation Life Time in [h]</i>	<i>Storage Time in [a]</i>	<i>Comment</i>
-40	480		Operation mission profile
23	1600		
40	5200		
80	640		
95	80		
105	50		
23		10	In production standby
85		5	
23		10	After market storage
95		5	
	8000	30	Sum

10 Abbreviations

Table 10-1: Abbreviations

Abbreviation	Explanation
ADC	Analog to Digital Converter
ASIC	Application Specific Integrated Circuit
ASSP	Application Specific Standard Product
ATG	Automatic Threshold Generation
CFG	Configuration
DAC	Digital to Analog Converter
EEPROM	Electrically Erasable Programmable Read-Only Memory
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EPD	Echo Peak Detection
ESD	Electrostatic discharge
ESR	Equivalent Serial Resistance
IC	Integrated Circuit
IO-IF	Input/Output Interface
JTAG	Joint Test Action Group
LSB	Least Significant Bit
MSB	Most Significant Bit
NFTG	Near-Field Threshold Generation
OV	Overvoltage
POR	Power-on Reset
RT	Ring time
STC	Sensitivity Time Control (also called dynamic gain control)
UV	Undervoltage

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11 General

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